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ABSTRACT

This tool is built to process mass inquiries in efficient manner and to save the effort of sales team. This tool will read the inquiry details like customer, material and required delivery date and based on historical quotation data available in the organization and by using data feed of third-party company.

Example: Haystack (data provider) which will increase the probability of quotation acceptance by the inquiring company or customer.

1. INTRODUCTION

An inquiry and quotation are the first documents in SAP SD which are a part of pre-sales business process. Inquiries and quotations help you to determine important sales related data and can be saved as documents. If the customer then places an order, this data can be accessed. Use this pre-sales information to plan and evaluate your marketing and sales strategies and as a basis for establishing long-term business relationships with your customers, for example by:

- Tracking lost sales
- Recording pre-sales data to help negotiate large contracts
- Selling goods and services to large organizations that require documentation of the entire process

Standard **SAP SD** inquiry normally contains:

- Customer and Material Information
- Pricing, be it Customer or Material specific
- Delivery dates and delivery quantities
- Information about shipment processing
- Information about billing
- In case of Inquiry, Quotation and Contract Validity Periods / Dates

2. LITERATURE SURVEY

2. LITERATURE SURVEY

2.1 SAP

E(Enterprise): It is a huge business organization.

R(Resources):

- Material: Material Management (MM)
- Money: FI(Finance)
- Manpower: HR (Human Resource)
- Machinery: PP/PM (Production Planning)/Plant Maintenance
- Marketing: SD (Sales and Distribution)
- Other Departments: CRM (Customer Relationship Management), SCM (Supply Chain Management)

P(Planning): It is a utilization of resources for maximum profits.

Definition: ERP (Enterprise Resource Planning): It is a single software with collection of departments such as SD (Sales and Distribution), MM (Material Management), HR, FI, etc so that this single software performs business activities of all the departments.

Different ERP packages:

1. JD Edwards: Accounting Software
2. PeopleSoft: HR
3. Siebel: CRM
4. Oracle Apps: Finance
5. SAP: All modules(departments)

SAP (Systems Applications and Products in data processing): SAP AG (ltd) was the company developed in by 5 ex-IBM employees in the year 1972 in small town called Walldron of country Germany.

SAP R/2: They developed the first product called as SAP R/2 in the year 1972. Here R is Real time and 2 is two-tier. But this product failed due to two reasons 1. SAP R/2 is costly and it runs of mainframes machines which are also costly. 2. It has 2-tier architecture which will decrease the business process.

1. SAP R/2 is costly and it runs of mainframes machines which are also costly.
2. It has 2-tier architecture which will decrease the business process.

SAP R/3:

SAP struggled for 20 years and developed another product called as SAP R/3 system. Here R is Real time and 3 is three-tier.

SAP R/3 Definition: It is an integrated standard software which contains collection of business modules such as SD, MM, HR, FI, PP, etc; so that it can perform the business activities or tasks of any departments.

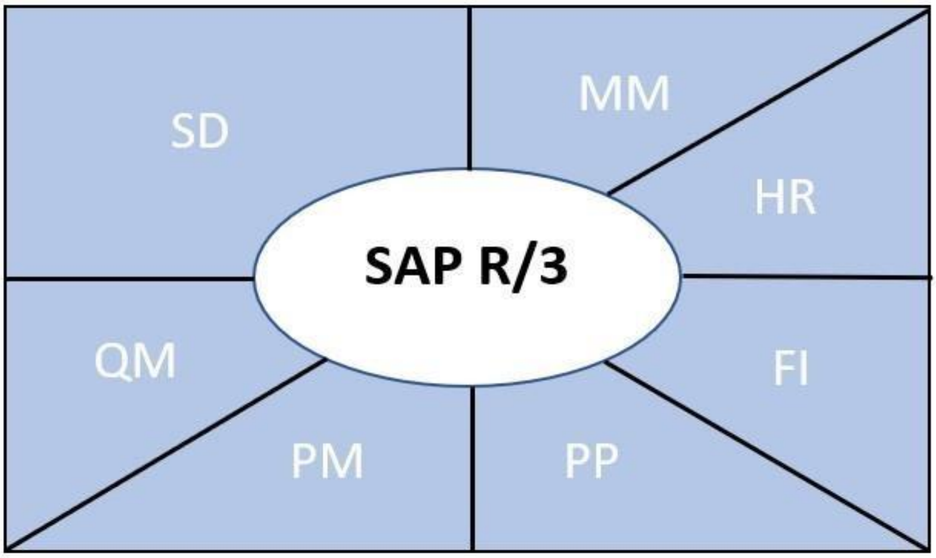


Fig2.1-SAP R/3 SYSTEM.

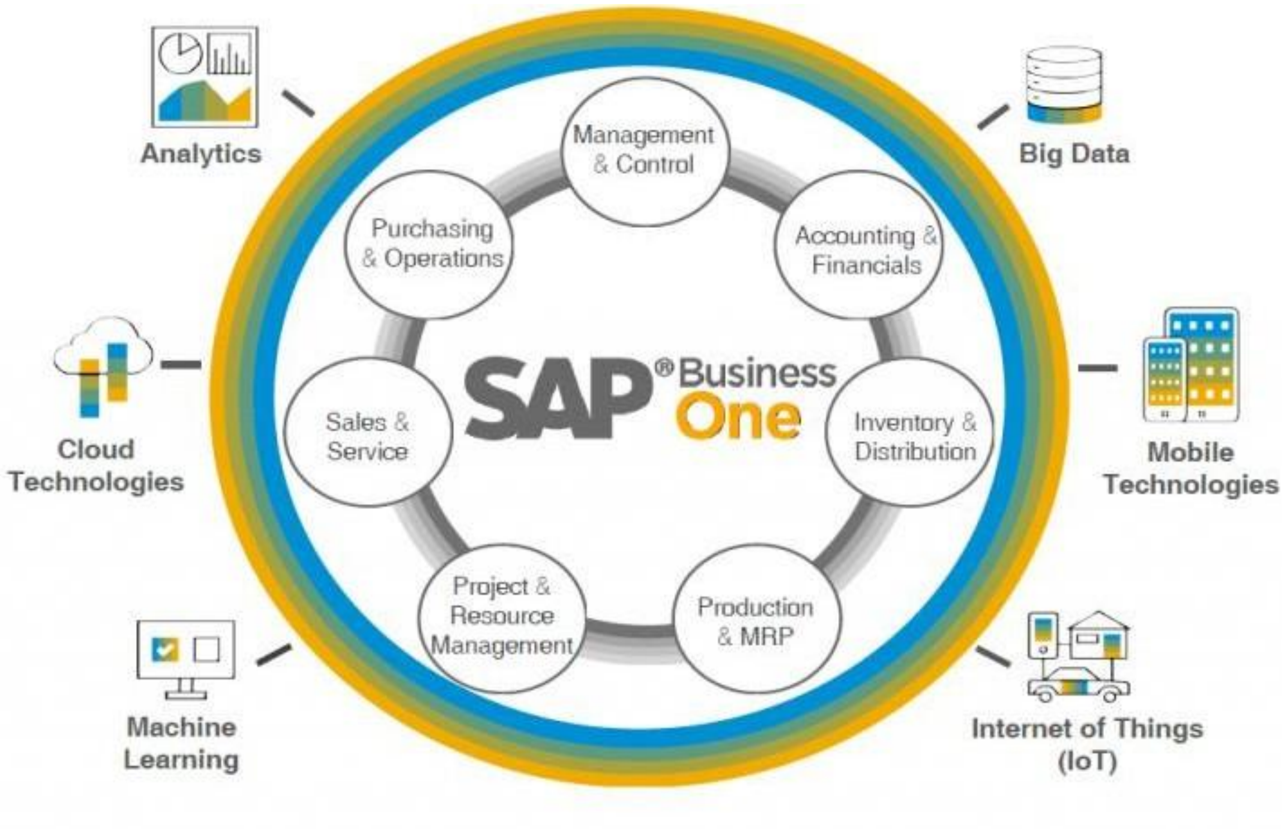
Main features of SAP R/3 System:

- Interface Communication), etc for exchangeof data with in remote systems.

- SAP R/3 is available in 17 different languages with their Currencies, Taxes formalities, Legal practice, Concerning HR, Import/Export Regulations.
- SAP R/3 system consists of more than 30 business application or function modules (such as SD, MM, PP, QM, PM, BW, FI, etc) with strong integration between the modules.
- It supports 24 hrs network between the different servers of R/3 system.
- It consists of more than 36,676 built-in database tables which business data, control and customizing database.
- It also consists of 18,000 built-in applications or servers for different business modules.
- SAP R/3 system is platform independent as well as database independent.That means it can run on any operating system such as UNIX, NT, 95, AS/400 and supports various RDBMS packages such as SQL server, Oracle, Informix, Sybase, DB2 ADABASE, etc.
- SAP R/3 system supports standard GUI's such as windows 95, NT, windows 3.1 and Macintosh.

  SAP R/3 system uses Standard Communication Protocol such as TCP/IP, CPIC (Common Programming

- Those features made almost every industry to implement SAP R/3 system such as
    - Oil and Gas.
    - Chemical.
    - Pharmaceutical.
    - Automotive.
    - Shipping, Aerospace and Train construction.
    - Healthcare and Hospitals.
    - Education Institutions and research centres.
    - Consulting and software.
    - Government, public Administration and Services.
    - Museums and Associations.
    - Building and Heavy constructions.
    - Raw materials, Mining and Agricultural fields.



2.1.1SAP BUSINESS DIAGRAM

Quotation in SAP SD

An inquiry and quotation are the first documents in SAP SD which are a part of pre-sales. In order to proceed with sales, i.e. quotation creation, inquiry and quotation are NOT mandatory documents which have to be maintained / entered in the system in order to proceed with the quotation/sales processing. Quotations serve as a legal document, i.e. you confirm the customer that you shall provide XYZ material on specific rate within the validity of the quotation.

Inquiries and quotations help you to determine important sales related data and can be saved as documents. If the customer then places an order, this data can be accessed. Use this pre-sales information to plan and evaluate your marketing and sales strategies and as a basis for establishing long-term business relationships with your customers, for example by:

- tracking lost sales
- recording pre-sales data to help negotiate large contracts
- selling goods and services to large organizations that require documentation of the entire process

Standard SAP SD quotation normally contains:

- Customer and material information
- Pricing, be it customer or material specific
- Delivery dates and delivery quantities
- Information about shipment processing
- Information about billing

2.2 DATA TYPES IN SAP

Data types: It describes the properties of variables such as data type and length.Data types does not occupy memory.

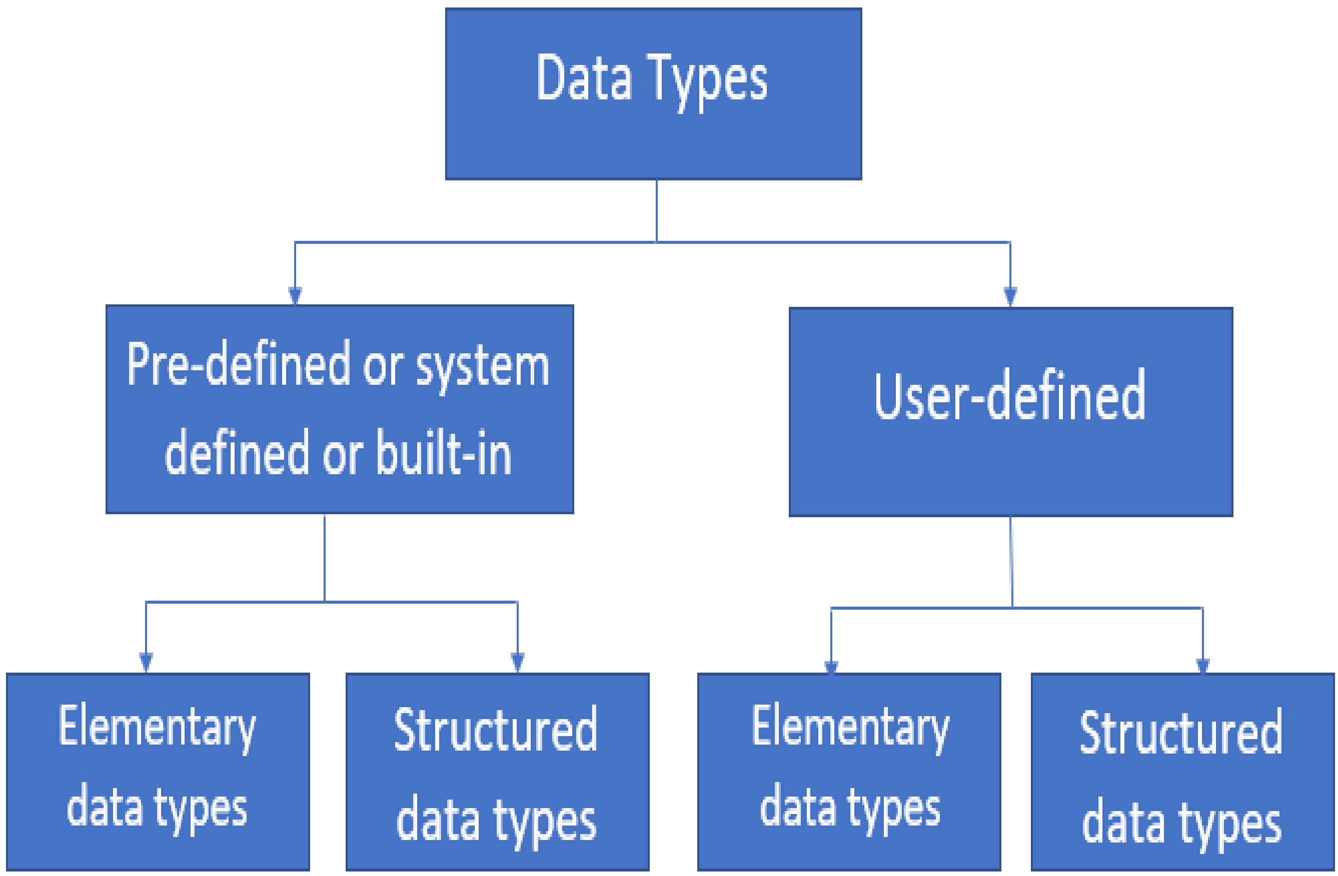


Fig-2.2 DATA TYPES

Pre-defined Elementary Data types: There are 10 pre-defined elementary data types.

- C(char),
- D(date),
- F(float),
- I(integer),

- N(Numeric),
- P(packed),
- T(time),
- X, STRINGAND XSTRING.

Data Type	Initial Size	Valid Size	Default value	Description
C	1	1-65,535	Space	Alphanumeric Text
D	8	8	‘00000000’	Yyyymmdd
F	8	8	0	Floating point (decimal values)
I	4		0	Integers
N	1	1-65,535	‘0……0’	Numeric text (only digits which are not used for calculation)
P	16	16	0	Packed Number
T	6	6	‘000000’	Hhmmss
X	1	1-65,535	‘x00…….’	Hexadecimals (binary data)

Fig-2.2.1 predefined Elementary data types.

- String – Variable Length or dynamic length - Alphanumeric text
- Xstring - Variable Length or dynamic length – Hexadecimals (binary data)
- If you don’t specify data type, then system will by default specify ‘C’ and length 1.

Pre-defined Structured data type:

```
Tables <database structure>
```

Example: tables Mara.

This statement creates implicit work area (group of fields) with the same nameand structure as database table.

User-defined data types:

'Types' keyword is used to create user-defined data types.

Syntax:

```
Types <datatype name>[(Len)] type <datatype>
[decimals].
```

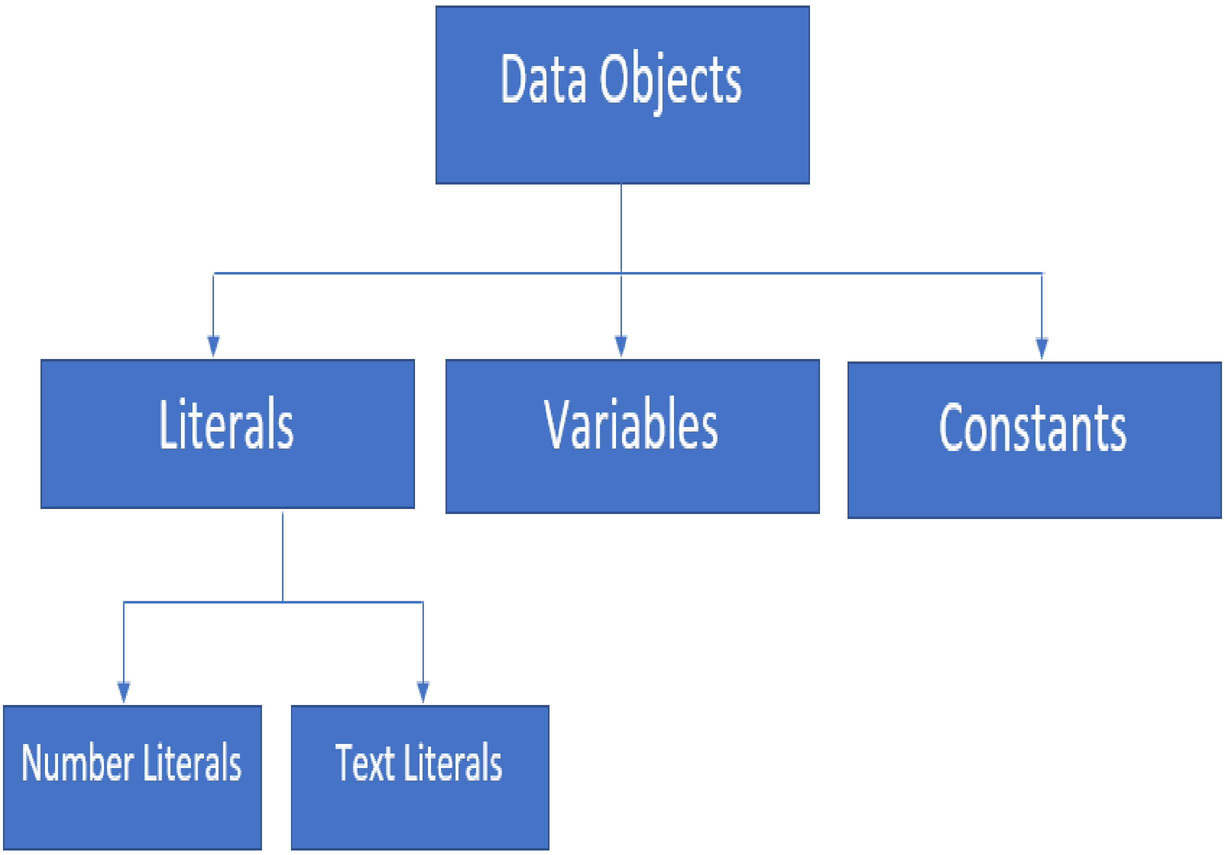


Fig-2.2.2 data objects:

- **Data objects:** are physical units which stores data during runtime or execution.

- **Literals**: They are fixed values.

    i. **Number Literal:** It contains digits.

    Ex: 95, 55, '75.5'

    ii. **Text Literal:** it contains alphanumeric text within single quotes.

    Ex: 'Abap is a programming language'.

- **Constants**: It contains value at the time of declaration only and the value remains unchanged throughout the execution of abap program.

'Constants' is the keyword to create constants.

Syntax:

Constants <cname>[(Len)] type/like <datatype> [decimals] <value> <val>.

- **Variable**: It stores a value or some data during runtime and value of variable keeps on changing during execution of program.

'Data' is the keyword to create variables.

Syntax:

```
Data <var_name> [(Len)] type/Like <datatype>[decimals]
[value <val>].
```

2.3 MASTER DATA INVOLVED:

- Master data is the core data that is used as a base for any transaction. If you are producing, transferring stock, selling, purchasing, doing physical inventory, whatever your activity may be, it requires certain master data to be maintained.
- The following master data is involved when creating an inquiry.

- This master data fetches relevant information and populates the relevant fields accordingly.

I. CUSTOMER MASTER DATA

- Name of the Customer
- Address
- Location
- Taxation details
- Geographical location according to company's Sales Geographical Structure
- Shipping details
- Billing details as in Terms of Payment
- Partners associated with the Customer I.e. Ship to Party, Billing, Payer etc.

II. MATERIAL MASTER DATA

- Material Description
- Unit of Measure (Stock Keeping Unit – SKU)
- Transportation Group
- Loading Group (fork lifter, manual handling etc.)
- Taxation information.

CUSTOMER MATERIAL INFO RECORDS

- Used to maintain customer relevant SKU codes which are mapped with the company's code

CONDITION MASTER (PRICING)

- Base price of the product or special price for a particular customer
- Discount(s) on the product or special discount offered to one customer
- Freight
- Taxes

Partners (Parties) Involved in SAP SD Inquiry Creation

- Sold-To Party: the entity that inquired the goods
- Ship-To Party: to whom goods are be delivered to
- Bill-To Party: the entity that is responsible to receive the bill
- Payer: the entity responsible for the payment.

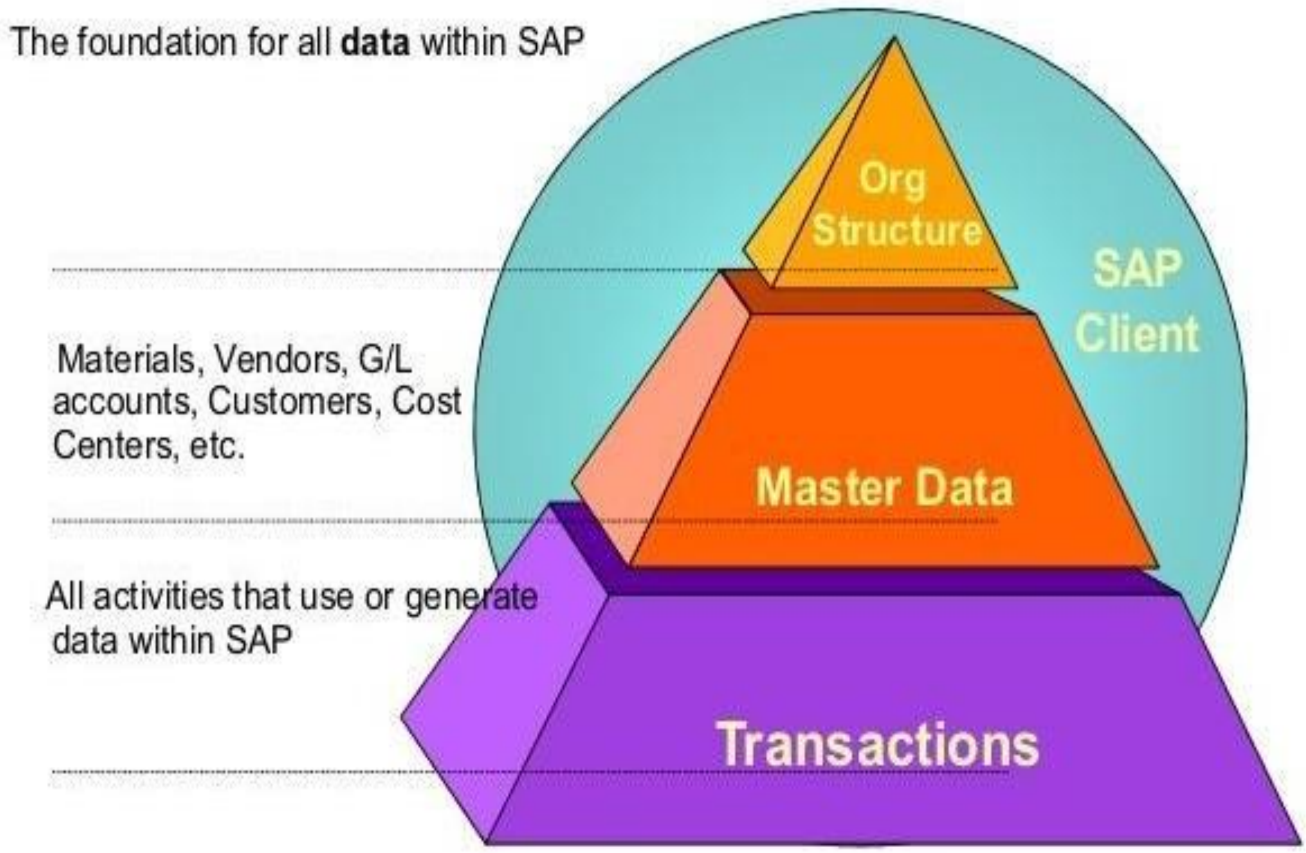


Fig- 2.3.1 Master data in SAP.

3. SYSTEM ANALYSIS

3. SYSTEM ANALYSIS.

1. Sales inquiries and sales quotations let you enter and store all the important, sales-related information you use during sales order processing. Sales inquiries and sales quotations that are not complex can be entered quickly in the initial entry screen. These can be entered from scratch or can be copied. For example, when the customer decides he wants a sales quotation, you can copy a previously entered sales inquiry directly into a sales quotation.

2. The sales inquiries and sales quotations entered in the system can be displayed and evaluated in a **list**. You can use selection criteria to limit the list, which will give you a more selective display and processing. For analysis purposes, you can list all the sales quotations you processed during the last six months and examine those that were rejected and for what reasons.

3. You can maintain a validity date in the sales inquiries and sales quotations to indicate by which time they should have been answered. The documents can then be monitored and evaluated according to this validity date, which then allows you to evaluate the sales inquiries and sales quotations on time. This way you are able to plan and implement the necessary subsequent activities according to the deadline.

4. Instead of a material number you can also enter a **text description of the material** if, for example you can’t assign the materials to a customer call straight away.

5. In one sales quotation for a material, you can offer the customer several options with differing pricing conditions and shipping times. If the quotation leads to a sales order, you can select an **alternative item according to the customer’s requirements**.

4. SYSTEM OVERVIEW AND REQUREMENTS

4. SYSTEM OVERVIEW AND REQUIREMENTS

4.1 Functional Requirements

1. User authentication: The system should require users to authenticate themselves before accessing the sales and inquiry quotation features.

2. Customer information management: The system should allow users to add, update, and view customer information, including contact details, billing information, and shipping information.

3. Product catalog management: The system should allow users to add, update, and view products and services, including pricing and availability information.

4. Quotation creation: The system should allow users to create quotations for customers, including details such as product or service name, quantity, price, discount, tax, and total amount.

5. Quotation management: The system should allow users to manage and track the status of quotations, including sending, accepting, rejecting, and revising quotations.

6. Reporting and analytics: The system should allow users to generate reports and analytics on sales and quotation data, including sales by product, sales by customer, and quotation conversion rates.

7. Integration with other systems: The system should allow integration with other systems such as CRM, ERP, and accounting systems to streamline data management and eliminate data duplication.

8. Customization and configuration: The system should allow users to customize and configure the system to meet their specific business requirements, including branding, workflows, and notifications.

9. Mobile access: The system should allow users to access the sales and inquiry quotation features from mobile devices such as smartphones and tablets.

10. Security and data privacy: The system should implement appropriate security measures and data privacy policies to protect sensitive customer and business data from unauthorized access or disclosure.

4.2 Non-Functional Requirements

1. Usability: The system should have a user-friendly interface and intuitive navigation to make it easy for users to create and manage quotations.

2. Performance: The system should be able to handle a large number of quotations and customers without any significant slowdowns or crashes.

3. Availability: The system should be available 24/7 to ensure that users can access and create quotations at any time.

4. Reliability: The system should be reliable and provide accurate information to users without any errors or inconsistencies.

5. Scalability: The system should be able to scale up or down depending on the size and complexity of the business without any significant impact on performance.

6. Security: The system should implement appropriate security measures to protect customer and business data from unauthorized access, including encryption and access controls.

7. Compatibility: The system should be compatible with different browsers, operating systems, and devices to ensure maximum usability and accessibility.

4.3 Software Requirements

Operating System	:	Windows 10 or MAC
OS.Platform	:	SAP LOGON
Programming Language	:	ABAP

4.4 Hardware Requirements

Processor	:	Intel core i3 and above.
Hard Disk	:	500 GB or above.
RAM	:	8 GB or above.
Internet	:	4 Mbps or above (Wireless).

5. SOFTWARE TOOLS ANDDESCRIPTION

5. SOFTWARE TOOLS AND DESCRIPTION

5.1 SAP ABAP

Abap/4 (Advance Business Application Programming Language):

- ABAP is a high level programming language created by SAP that helps large corporations to customize the SAP ERP.
- ABAP can help customize workflows for financial accounting, materials management, asset management, and all other modules of SAP.
- SAP's current development platform NetWeaver also supports both ABAP and Java. The ABAP full form in SAP is Advanced Business Application.

4: 4^{Th} Generation.

Definition: ABAP is a programming language to develop business application.

Application: it a set of programs which performs a business activity or task.

SAP provided different program types:

1. Program type 1: **Executable program (Report)**
2. Program type M: **Module pool program**
3. Program type S: **Subroutine pool program**
4. Program type I: **Include program**
5. Program type K: **Classes**
6. Function Group: **F**

Executable program: It is an abap program which contains set of statements that are executed directly which displays some output or result.

Non-executable program: It is an abap program which contains a set of statements but are not executed directly, to execute this type of program it should be called from an executable program.

Example: Subroutine pool program: A program pool(collection) of subroutine (set of statements) which are not executed directly, to execute this program pool it should be called from a different program i.e., executable program.

ABAP development workbench (or) abap workbench:

It is an integrated programming environment which contains number of tools to develop abap application.

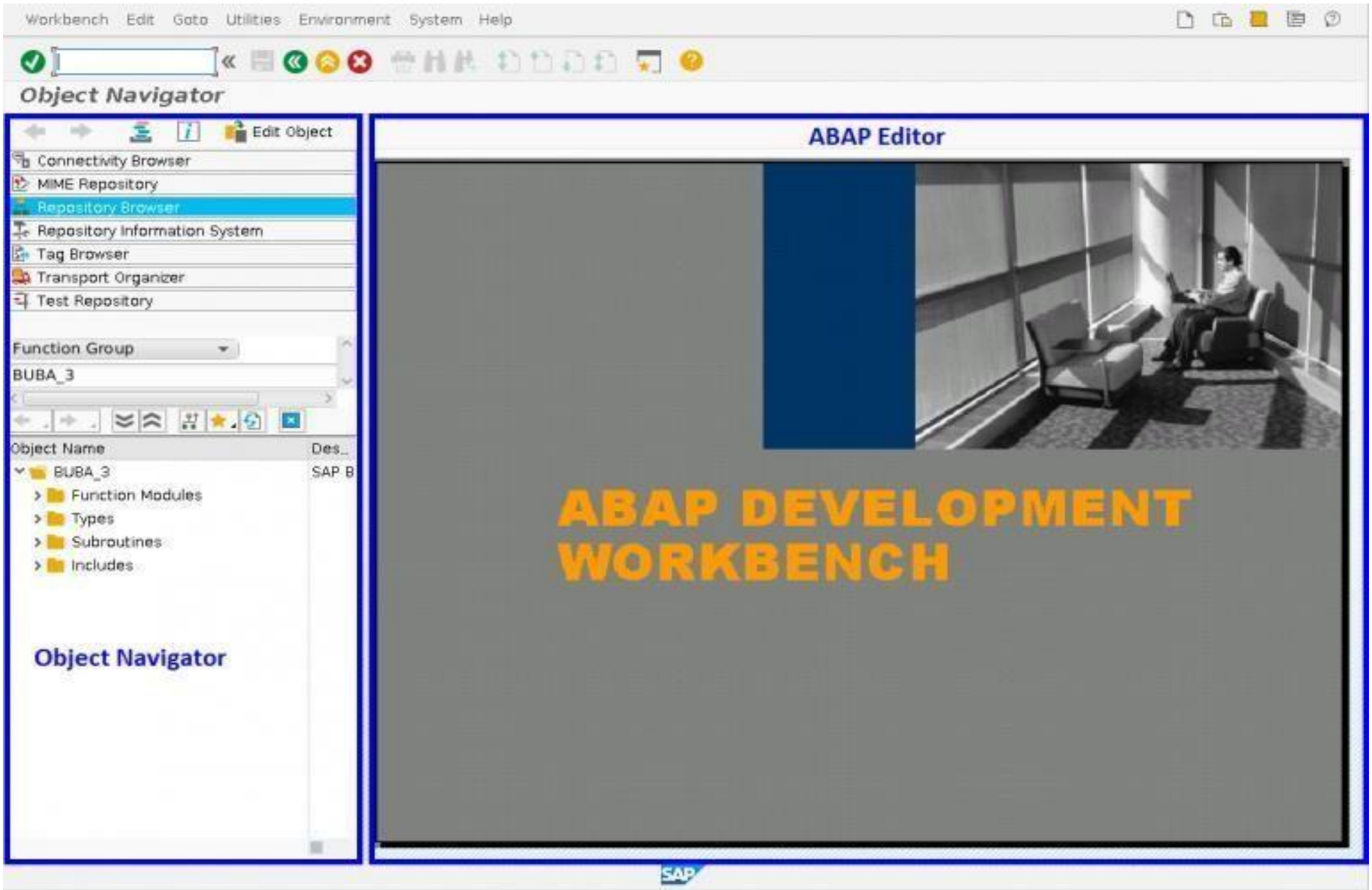


Fig:5.1 ABAP development workbench.

Tools are:

1. **<u>Abap Editor:</u>** Tcode (Transaction Code): **SE38**: It is used to create abapprograms.
2. **<u>Data Dictionary (or) Abap Dictionary</u>**: Tcode: **SE11:** It is used to createdatabase tables.
3. **<u>Screen Painter</u>**: Tcode: **SE51**: It is used to design screens.
4. **<u>Menu Painter</u>**: Tcode**: SE41**: It is used to create GUI (Graphical User Interface) which contains menu bar, application tool bar, standard tool bar which contains pushbutton, icons, etc; so that user can more easilyinteract with screen.
5. **<u>Function Builder:</u>** Tcode: **SE37**: It is used to create functions.
6. **<u>Object Browser or Object Navigator</u>**: Tcode: **SE80**: It is used to create a complete application or objects.

There are two ways to navigate to any of the tools:

1. Through Navigation:

   SAP Menu->Tools->Abap workbench->Development->Abap Editor(SE38).
2. Provide the Transaction Code (Tcode) in the command field of standardtool bar and hit enter.
3. : /N-> back to sap easy access (main screen)
4. /O->it creates new session (R/3 window).

5.2 LIBRARIES:

1. British Council Library - They have a vast collection of technical books and resources, including books on SAP ABAP programming.

2. Hyderabad Public Library - They have a dedicated section for technical books, and you may find books on SAP ABAP programming.

INSTALLING SOFTWARE;

To install SAP Logon, please follow these steps:

1. Download the SAP Logon installer from the SAP support portal. You will need an SAP user account to access the portal.
2. Run the installer and select the desired installation language.
3. Accept the license agreement and click "Next".
4. Select the installation type. You can choose between a complete installation or a custom installation where you can select the components to install.
5. Specify the destination folder where SAP Logon should be installed and click "Next".
6. Choose the components that you want to install and click "Next".
7. Review the summary of your installation choices and click "Install".
8. Wait for the installation to complete.
9. Once the installation is complete, click "Finish" to close the installer.

After the installation, you can launch SAP Logon and add connections to SAP systems that you want to access.

6. OBJECTIVE

6. OBJECTIVE

Sales inquiries and sales quotations let you enter and store all the important, sales-related information you use during sales order processing. Sales inquiries and sales quotations that are not complex can be entered quickly in the initial entry screen. These can be entered from scratch or can be copied. For example, when the customer decides he wants a sales quotation, you can copy a previously entered sales inquiry directly into a sales quotation.

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7.SYSTEM DESIGN

7. SYSTEM DESIGN

7.1 System Architecture.

The architecture of the proposed system is as displayed in the figure below. The major components of the architecture are as follows: Master data, quotation request from customer, company analyze, data on which the document is created, customer group and price list.

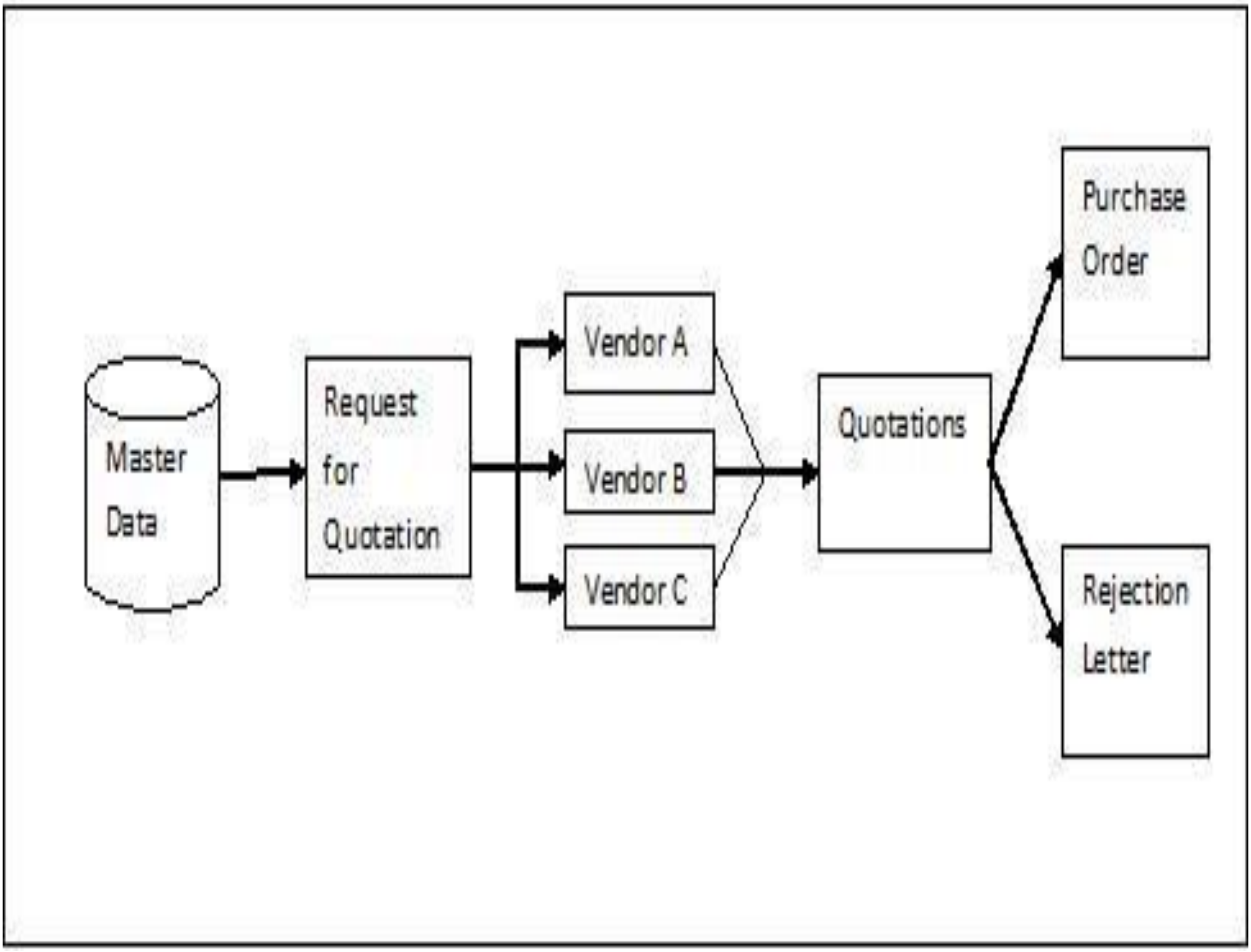


Fig-7.1 system architecture

- A sales inquiry comprises one or more items that contain the quantity of a material or service that the customer asked for.

- The total quantity of an inquiry item can be subdivided between the schedule lines in different amounts and relevant delivery dates.

- Items can be structured in a hierarchy so that you can differentiate between batches or explode any combinations of materials (bills of material).

- The valid conditions for an item are listed in the item conditions and can be derived individually from the conditions for the whole document.

7.2 Introduction to UML

The Unified Modeling Language allows the software engineer to express an analysis model using the modeling notation that is governed by a set of syntactic, semantic and pragmatic rules. A UML system is represented using five different views that describe the system from distinctly different perspective. Each view is defined by a set of diagrams, which is as follows:

1. User Model View

   This view represents the system from the users' perspective. The analysis representation describes a usage scenario from the end-users' perspective.

2. Structural Model View

   In this model, the data and functionality are arrived from inside the system. This model view models the static structures.

3. Behavioral Model View

   It represents the dynamic of behavioral as parts of the system, depicting he interactions of collection between various structural elements described in the user model and structural model view.

4. Implementation Model View

   In this view, the structural and behavioral as parts of the system are represented as they are to bebuilt.

5. Environmental Model View

   In this view, the structural and behavioral aspects of the environment in which the system is to be implemented are represented.

UML DIAGRAMS

7.2.1 Use-Case Diagram

To model a system, the most important aspect is to capture the dynamic behavior. To clarify a bit in details, dynamic behavior means the behavior of the system when it is running/operating.

So only static behavior is not sufficient to model a system rather dynamic behavior is more important than static behavior. In UML there are five diagrams available to model dynamic nature and use case diagram is one of them. Now as we have to discuss that the use case diagram is dynamic in nature there should be some internal or external factors for making the interaction.

These internal and external agents are known as actors. So, use case diagrams are consisting of actors, use cases and their relationships. The diagram is used to model the system/subsystem of an application. A single use case diagram captures a particular functionality of a system. So, to model theentire system numbers of use case diagrams are used.

Use case diagrams are used to gather the requirements of a system including internal and external influences. These requirements are mostly design requirements. So, when a system is analyzed to gatherits functionalities use cases are prepared and actors are identified. In brief, the purposes of use case diagrams can be as follows:

a. Used to gather requirements of a system.
b. Used to get an outside view of a system.
c. Identify external and internal factors influencing the system.
d. Show the interacting among the requirements reactors.

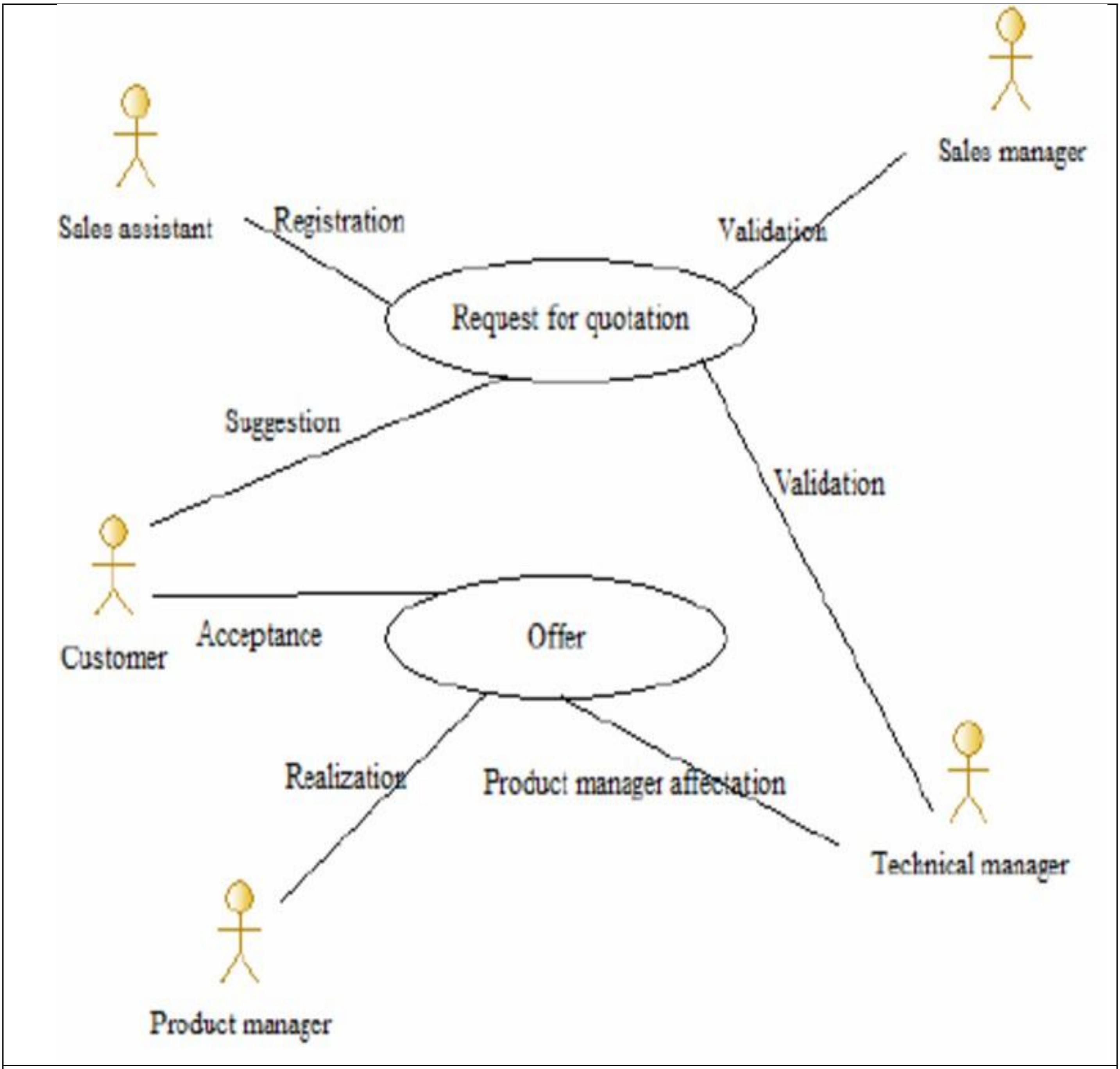


Fig 7.2.1 – Use Case Diagram

7.2.2 Sequence Diagram

Sequence diagrams describe interactions among classes in terms of an exchange of messages over time. They're also called event diagrams. A sequence diagram is a good way to visualize and validate various runtime scenarios. These can help to predict how a system will behave and to discoverresponsibilities a class may need to have in the process of modelling anew system.

The aim of a sequence diagram is to define event sequences, which would have a desired outcome. The focus is more on the order in which messages occur than on the message per se. However, the majority of sequence diagrams will communicate what messages are sent and the order in which they tend to occur.

Class Roles or Participants

Class roles describe the way an object will behave in context. Use the UML object symbol to illustrate class roles, but don't list object attributes.

Activation or Execution Occurrence

Activation boxes represent the time an object needs to complete a task. When an object is busy executing a process or waiting for a reply message, use a thin grey rectangle placed vertically on its lifeline.

Messages

Messages are arrows that represent communication between objects. Use half-arrowed lines to represent asynchronous messages.

Asynchronous messages are sent from an object that will not wait for are response from the receiver before continuing its tasks.

Lifelines

Lifelines are vertical dashed lines that indicate the object's presence over time.

Loops

A repetition or loop within a sequence diagram is depicted as a rectangle. Place the condition for exiting the loop at the bottom left corner in square brackets [].

When modelling object interactions, there will be times when a condition must be met for a message to be sent to an object. Guards are conditions that need to be used throughout UML diagrams to control flow.

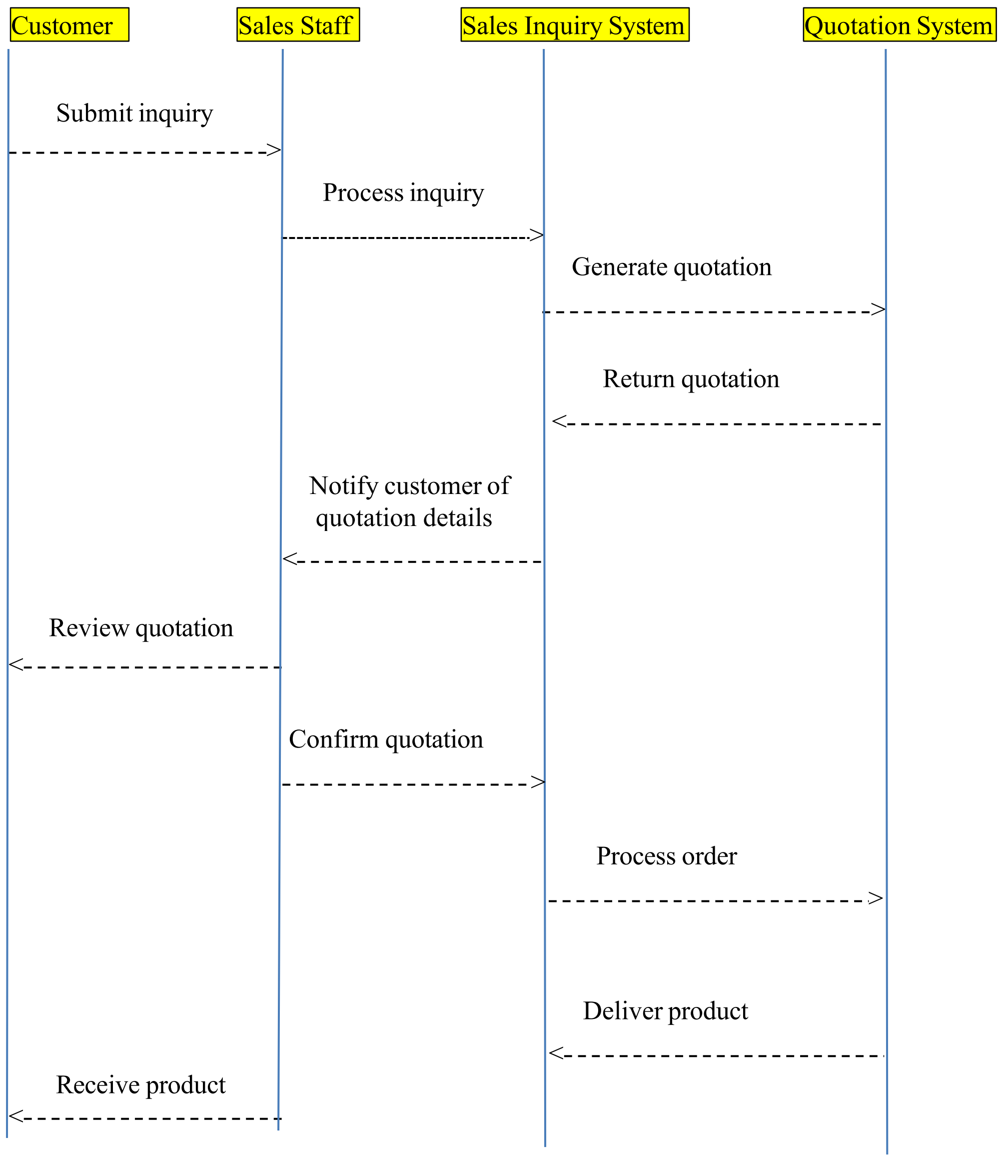


Fig 7.2.2 – Sequence Diagram

7.2.3 Class Diagram

Class diagrams are the main building blocks of every object-oriented method. The class diagram can be used to show the classes, relationships, interface, association, and collaboration. UML is standardized in class diagrams. Since classes are the building block of an application that is based on OOPs, so as the class diagram has appropriate structure to represent the classes, inheritance,relationships, and everything that OOPs have in its context. It describes various kinds of objects andthe static relationship in between them.

The main purpose to use class diagrams are:

This is the only UML which can appropriately depict various aspects of Obscenest.

1. Proper design and analysis of application can be faster and efficient.
2. It is base for deployment and component diagram.

Each class is represented by a rectangle having a subdivision of three compartments name, attributes and operation.

Dependency:

A dependency is a semantic relationship between two or more classes where change in one class cause changes in another class. It forms a weaker relationship.

Generalization:

A generalization is a relationship between a parent class (superclass) and a child class (subclass). In this, the child class is inherited from the parent class.

Association:

It describes a static or physical connection between two or more objects. It depicts how many objects are there in the relationship.

Multiplicity:

It defines a specific range of allowable instances of attributes. In case if a range is not specified, one is considered as a default multiplicity.

Aggregation:

An aggregation is a subset of association, which represents has a relationship. It is more specific than association. It defines a part-whole or part-of relationship. In this kind of relationship, the child class can exist independently of its parent class.

Composition:

The composition is a subset of aggregation. It portrays the dependency between the parent and its child, which means if one part is deleted, then the other part also gets discarded. It represents a whole-part relationship.

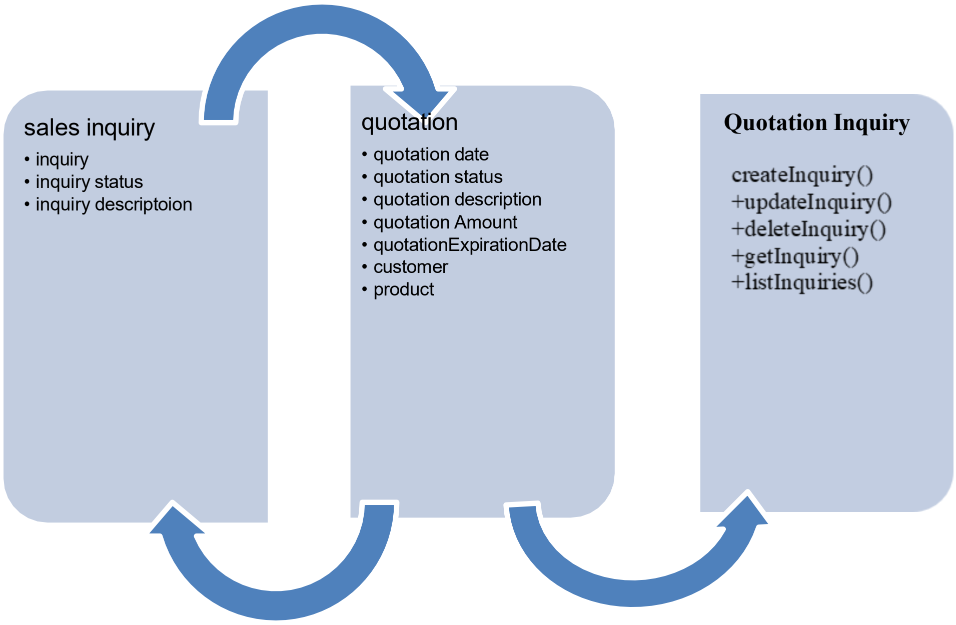


Fig 7.2.3-Class Diagram

8. IMPLEMENTATION

8. IMPLEMENTATION

8.1 Navigation steps to Login into SAP R/3 system:

1. Double click on Sap logon pad from desktop or frontend or presentation server.
2. Opens a screen, select 'Dev' Server or system and click on 'Logon' button.
3. Opens a screen, which contains 4 fields such as:

   Client: It is a number and is a highest unit in an organization. It is a groupof users who has similar rights.

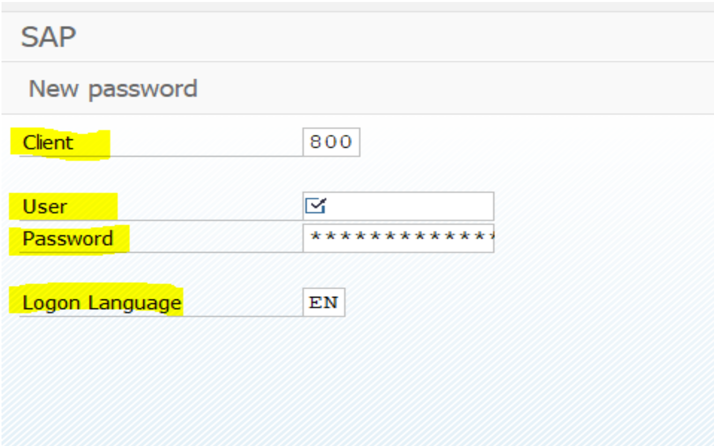
   

   Fig-8.1 sap login

   User: It is an SAP user identification and is client specific that means a userbelonging to a particular client is valid to that client only.

   Password: Given by Admin(basis).

   Language: Optional

   Provide: **client:** 800

User:	mohammad
Password:	india123

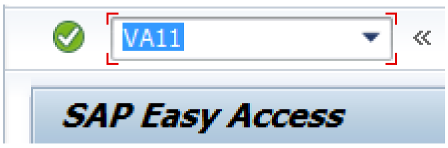
   And hit enter.

4. Opens a screen which is the main screen of SAP R/3 system called as Session or R/3 window which contains following components:

➢ Analyze Inquiry Transaction

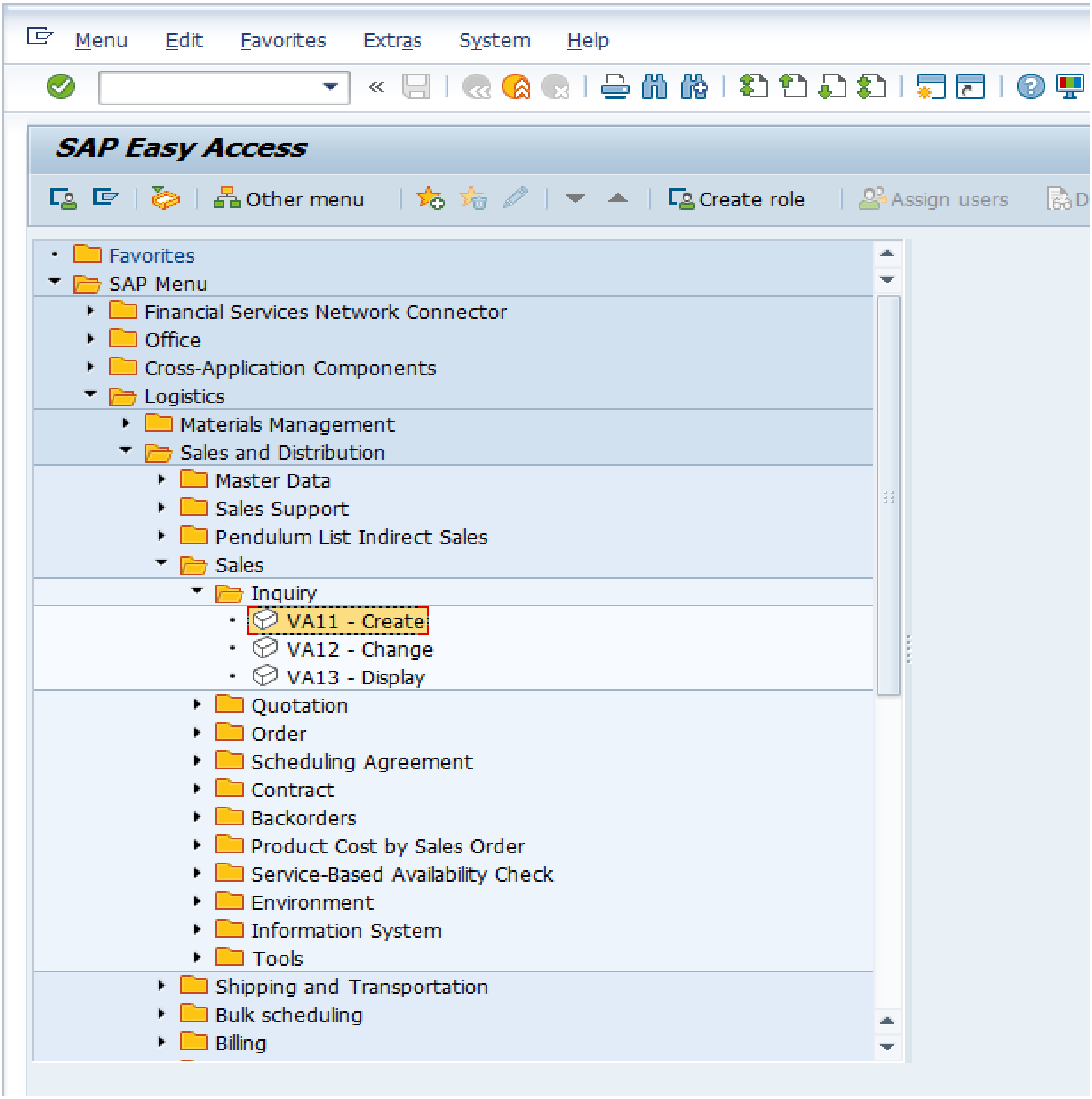
There are two options:

- Enter **VA11** in Transaction Command Field for creating Inquiry or



Enter **VA11** Transaction Code.

- Navigate through 'Tree'.



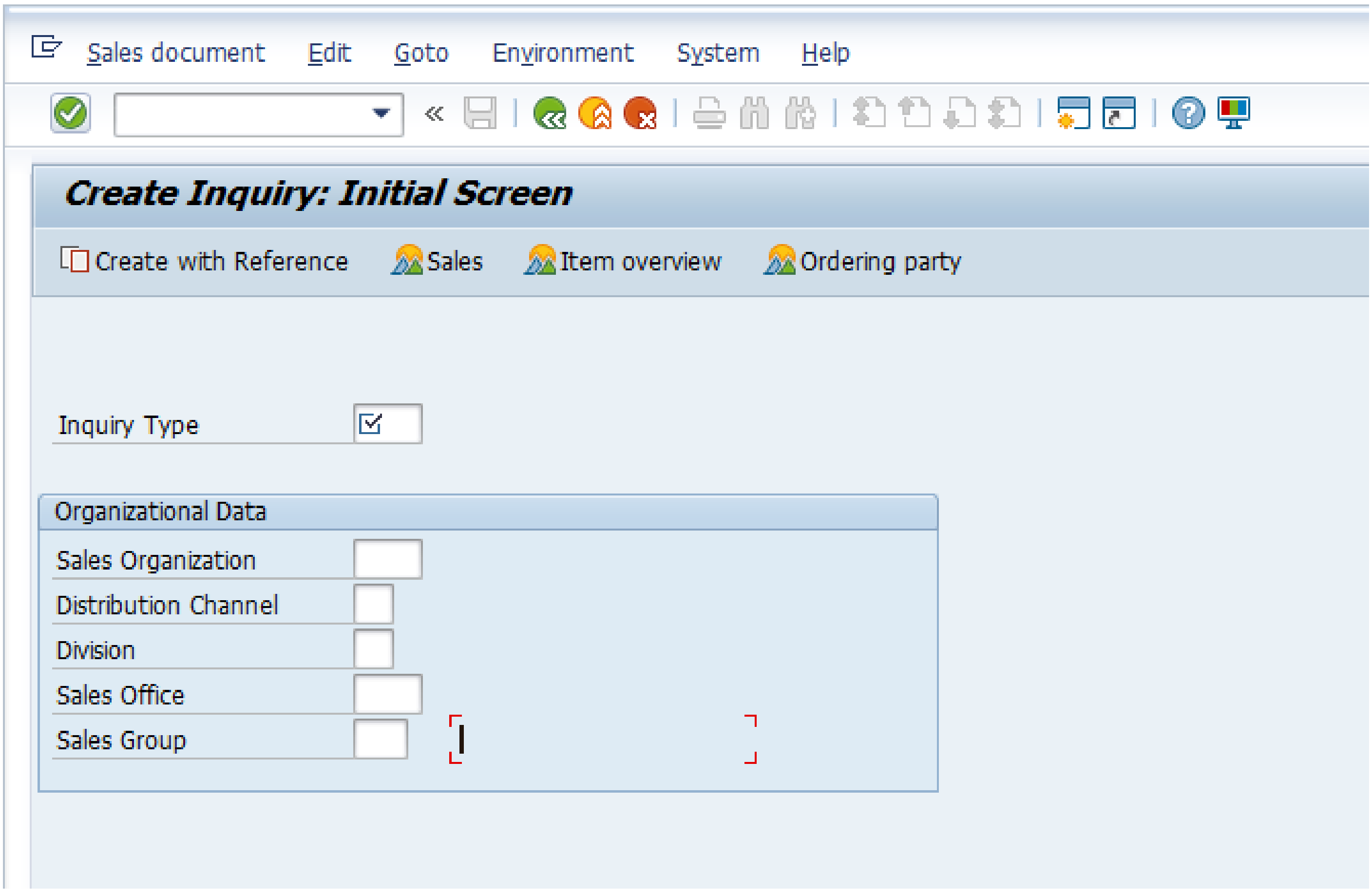
VA11 Transaction for Creation of Inquiry in SAP Menu.

- **Create SAP SD Inquiry**

Once you have entered the Inquiry: Creation Initial screen, you have to select relevant Inquiry Type. Each Inquiry type has its own functionality and viewing based on the configuration done at backend.

- **Select Relevant Inquiry Type(s)**

Standard Inquiry Type of Inquiry is: **IN**

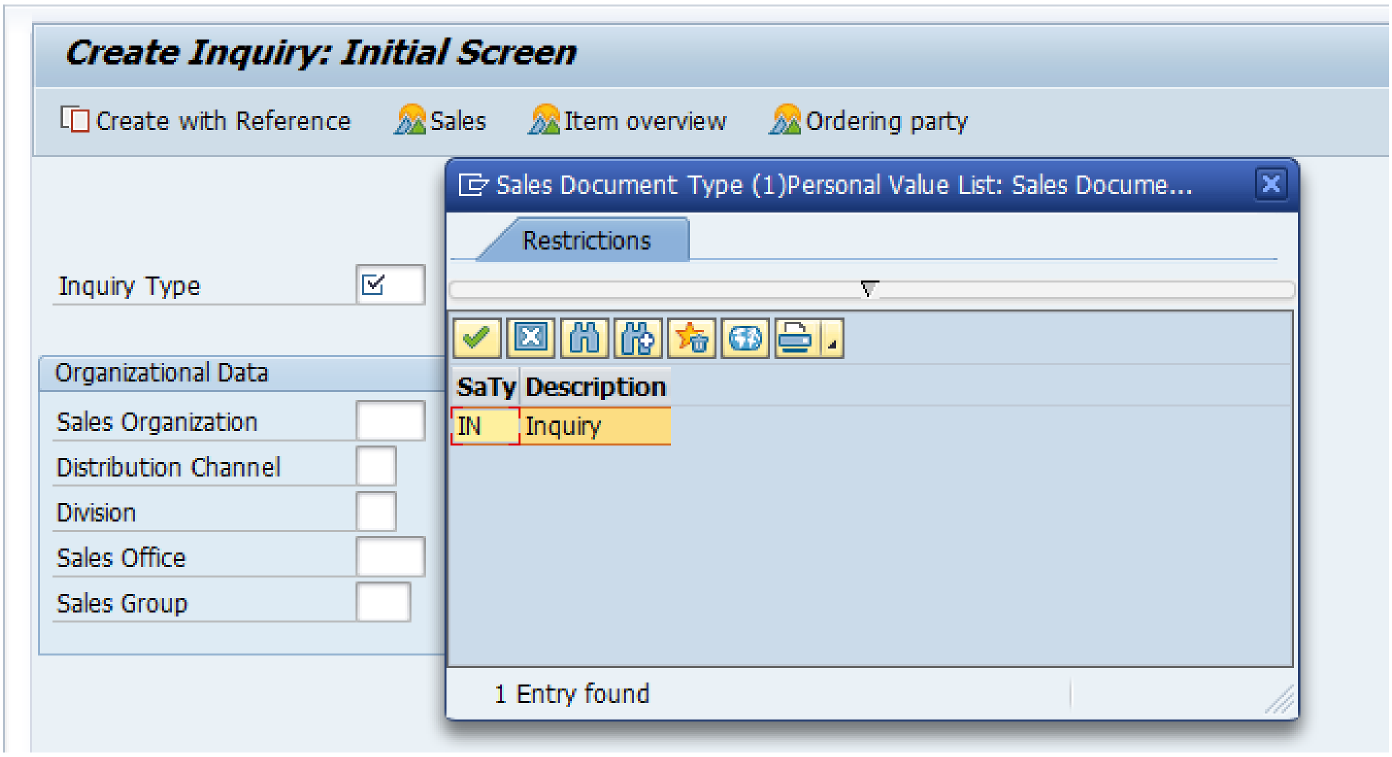


8.1.1 Create Inquiry – Initial Screen

- Please note, any field having a 'tick mark' 

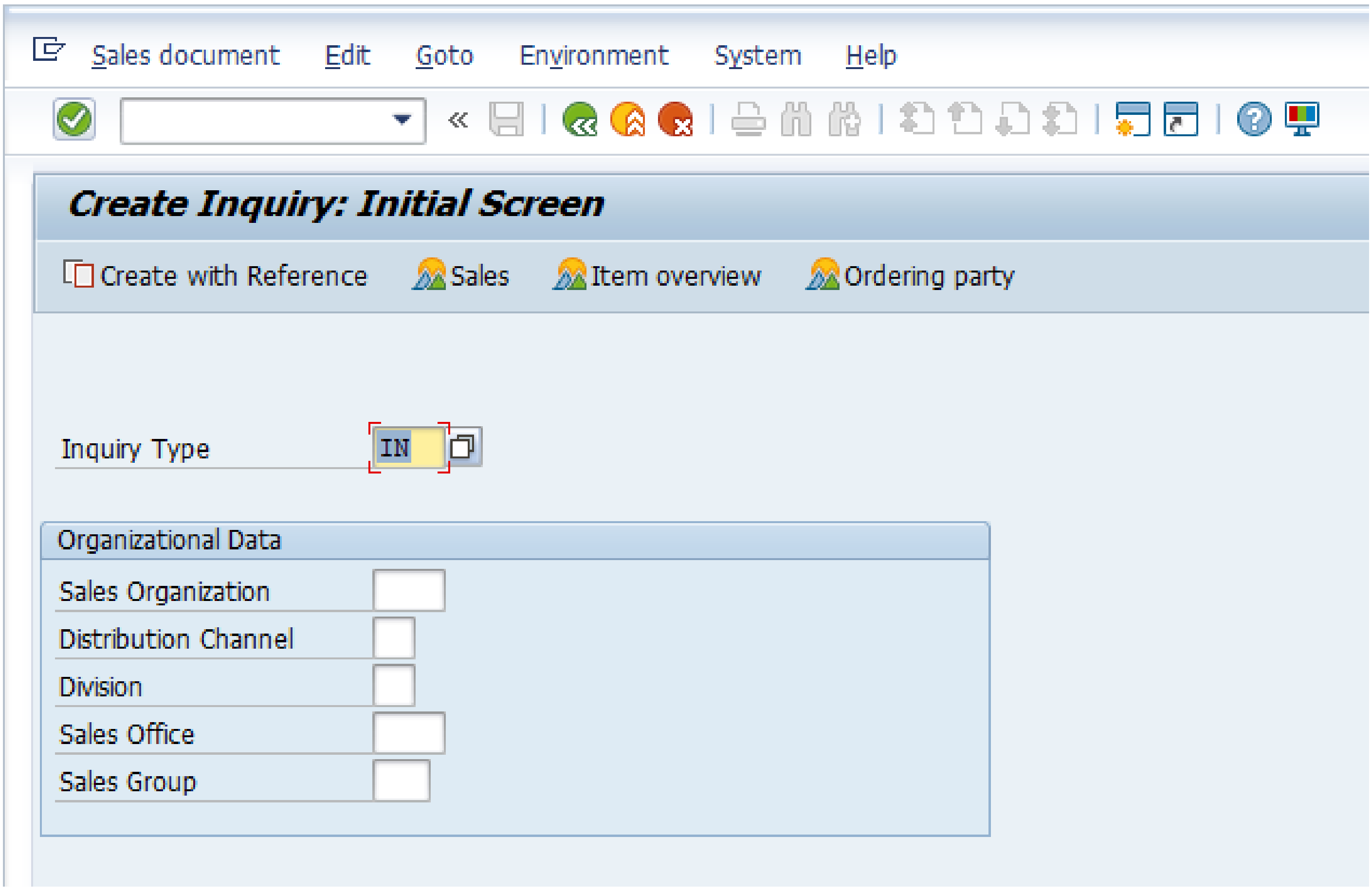
means its mandatory and it has to be populated, else the SAP system would not allow you to proceed further.

- In Inquiry, to opt for a required Inquiry type press *F4* button on the respective field and it will show all LoVs (List of Values).



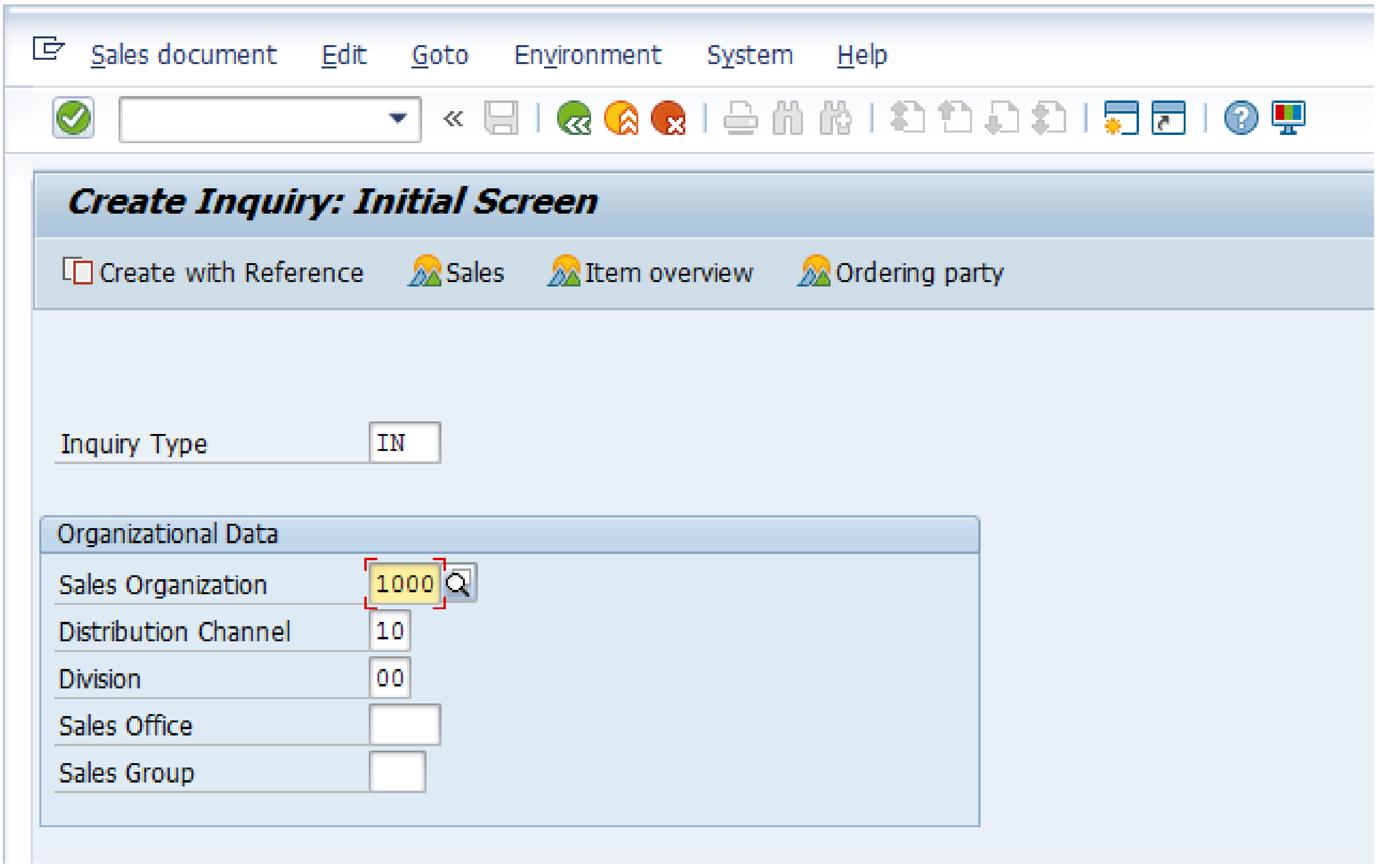
List with Inquiry Types

Select the relevant Inquiry type to proceed to the next step.



Create a Standard Inquiry

Populate Sales Organization with your relevant Sales Organization, Distribution Channel and Division. Click *Enter* button.



Fill in Sales Area Data

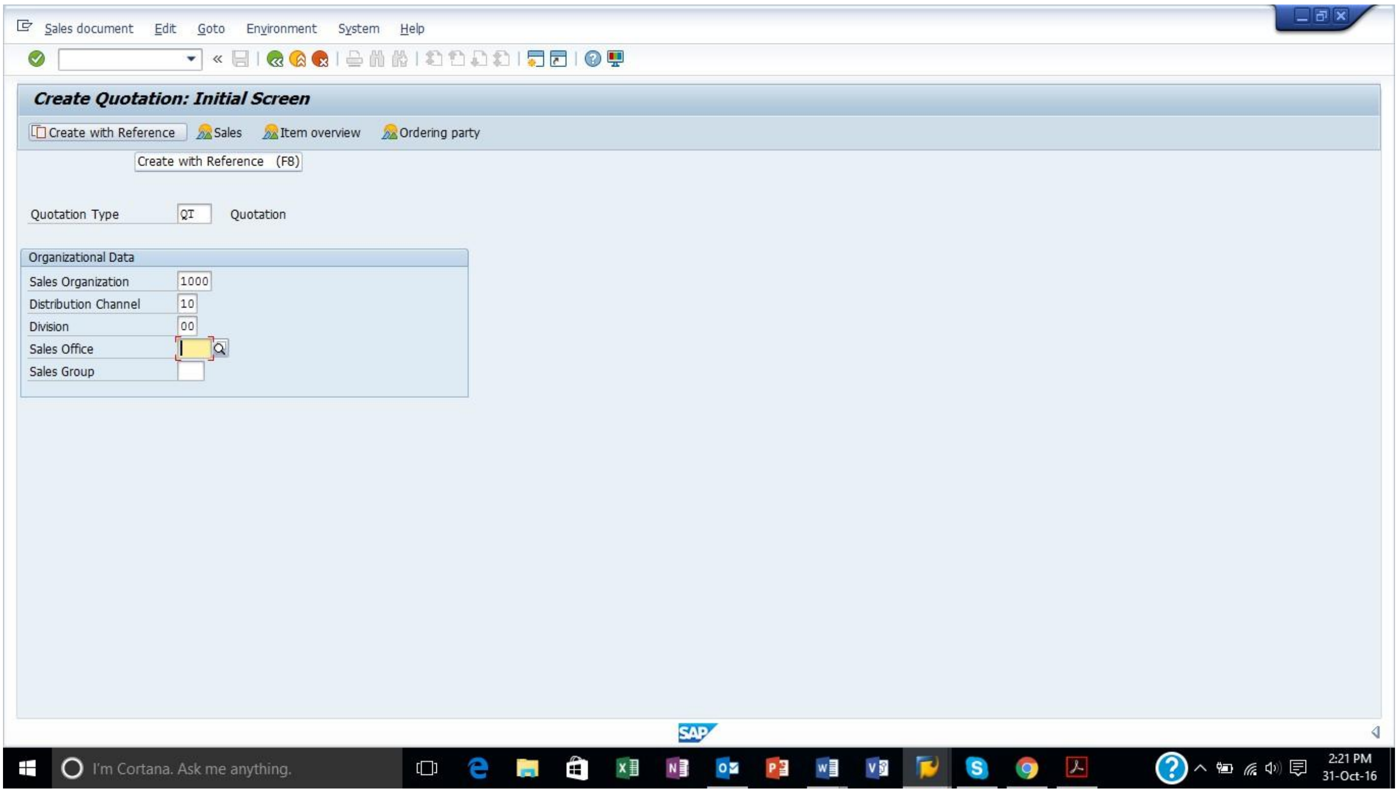
Hint: if you populate your relevant Sales Organization, Distribution Channel and Division first and then opt for the relevant Inquiry type, only Inquiry types relevant for the selected Sales Organization, Distribution Channel and Division will be displayed in LoVs.

Create Quotation with Reference to Inquiry

If you want to create a quotation with reference to an inquiry,

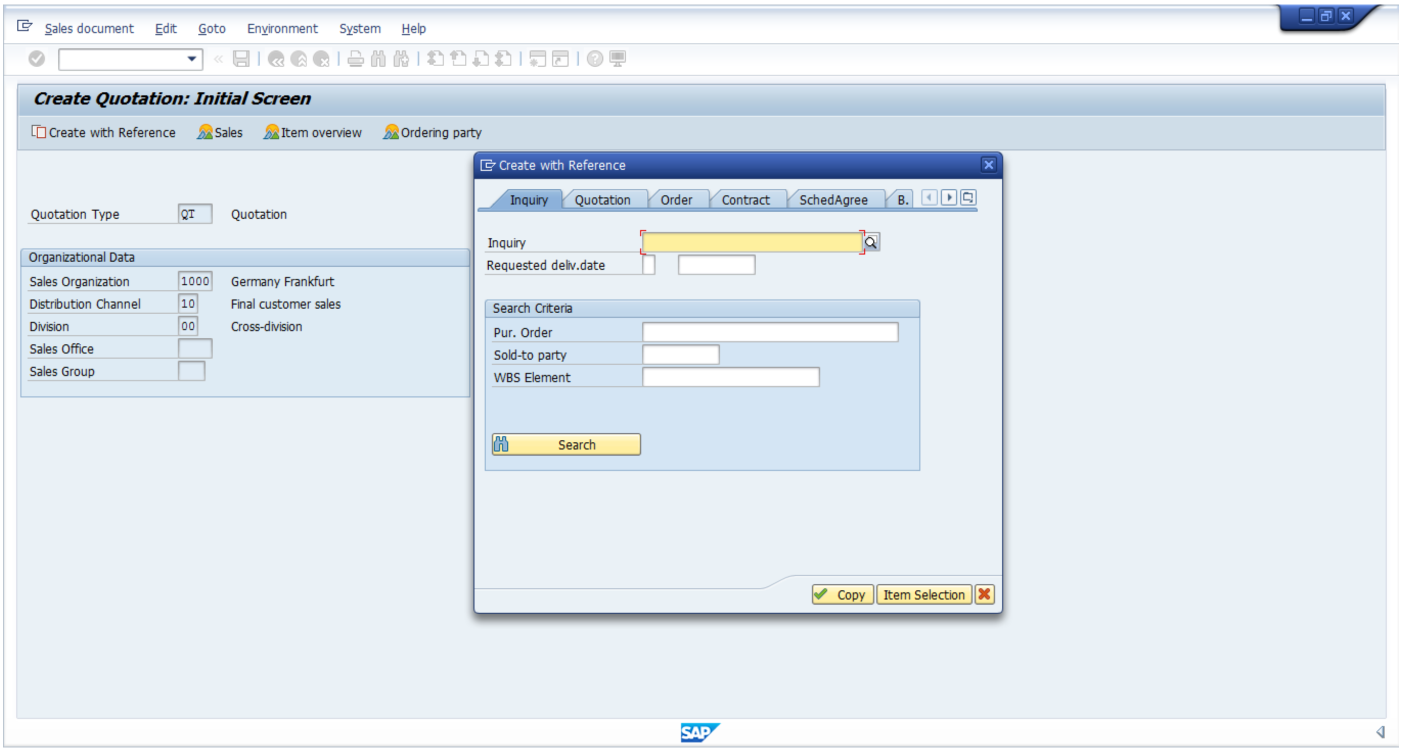
press: 

or *F8* button.

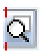


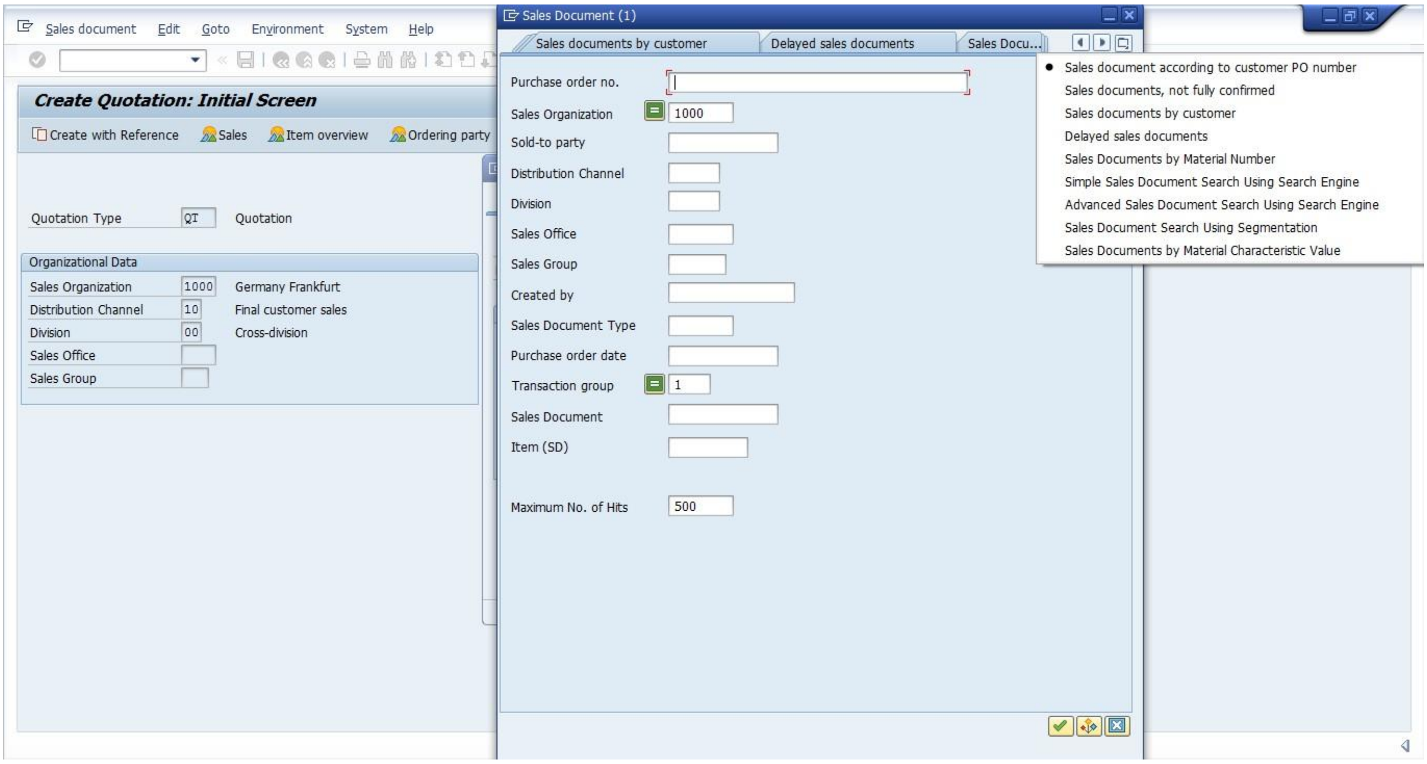
Create a Quotation with Reference to an Inquiry

Once it is pressed, a dialog box will appear and you need to select the relevant tab.



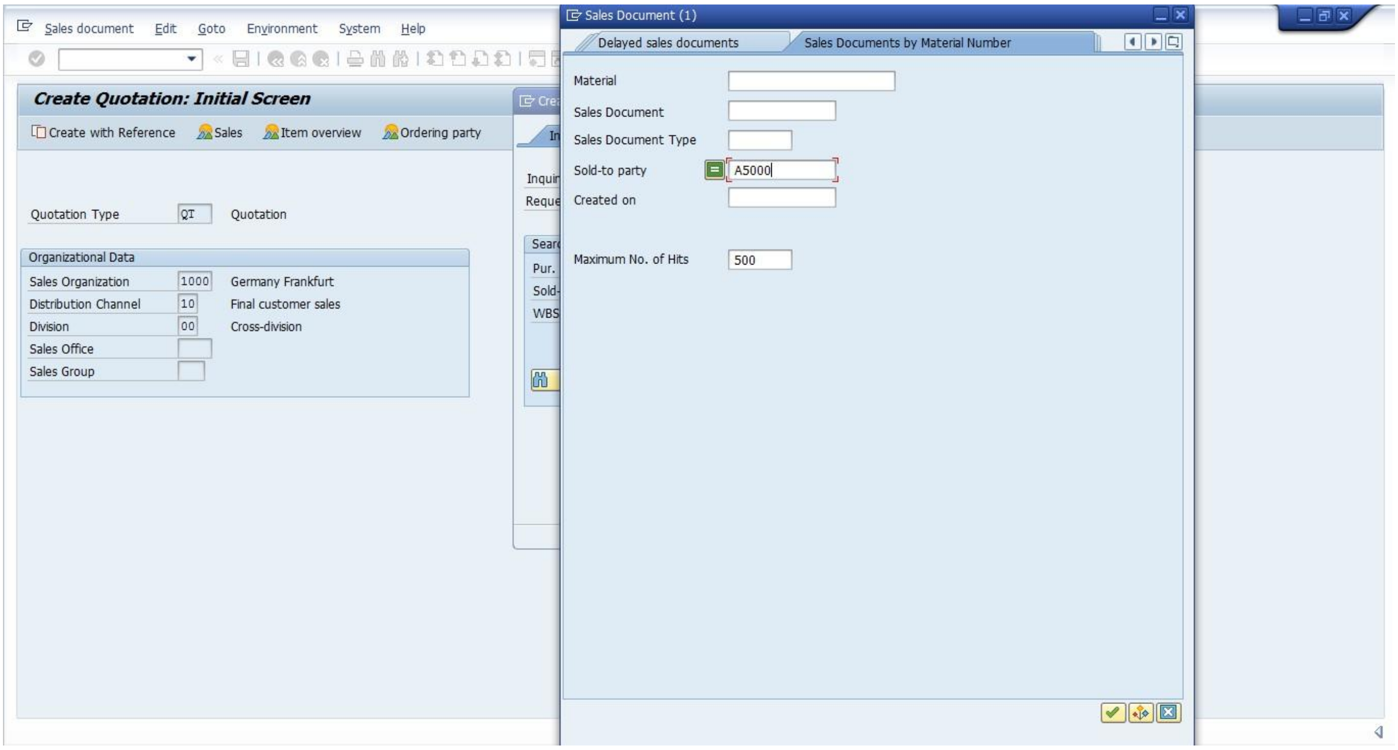
Create with Reference to Inquiry

Enter the document (in our case inquiry) number. If you know the number, enter the number directly or press *F4* or  button to search for the desired inquiry document.



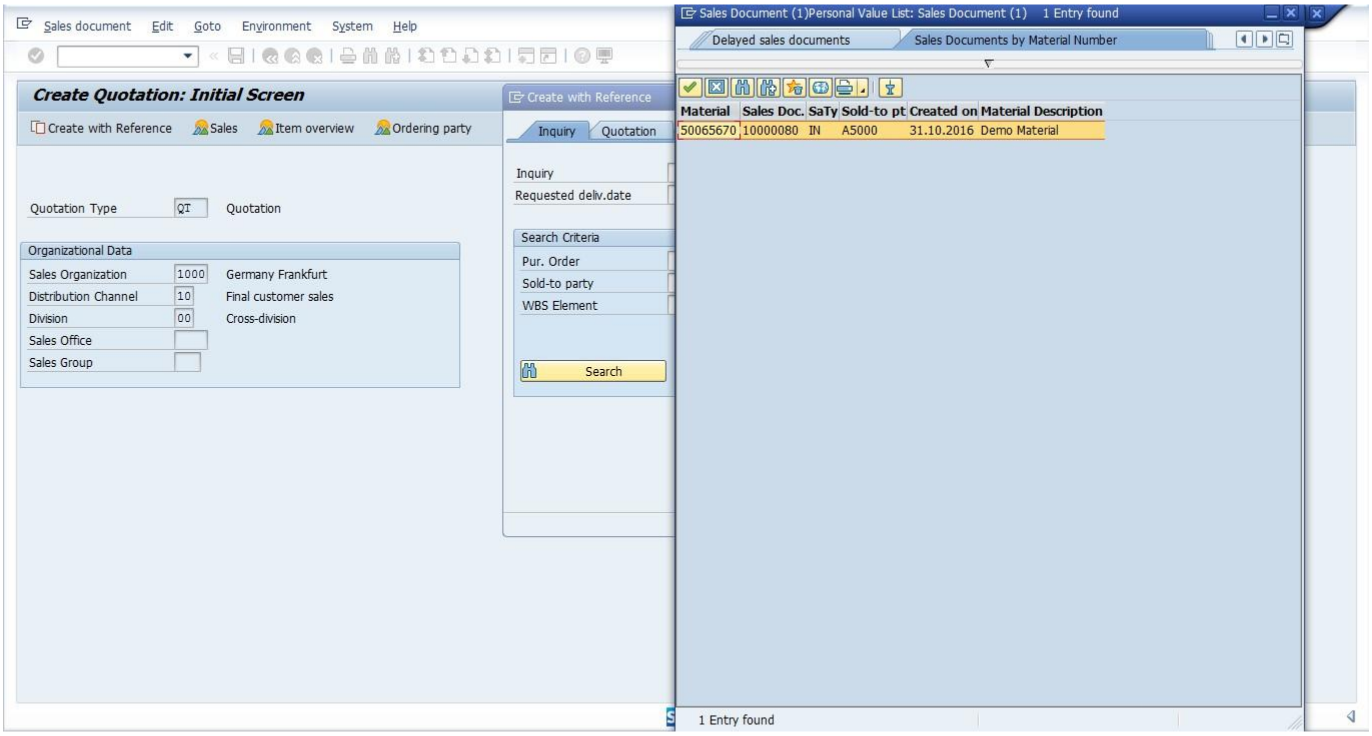
Search for Inquiry Document Number if You Don't Know It

For example, enter sold-to party code to search for desired inquiry by customer.



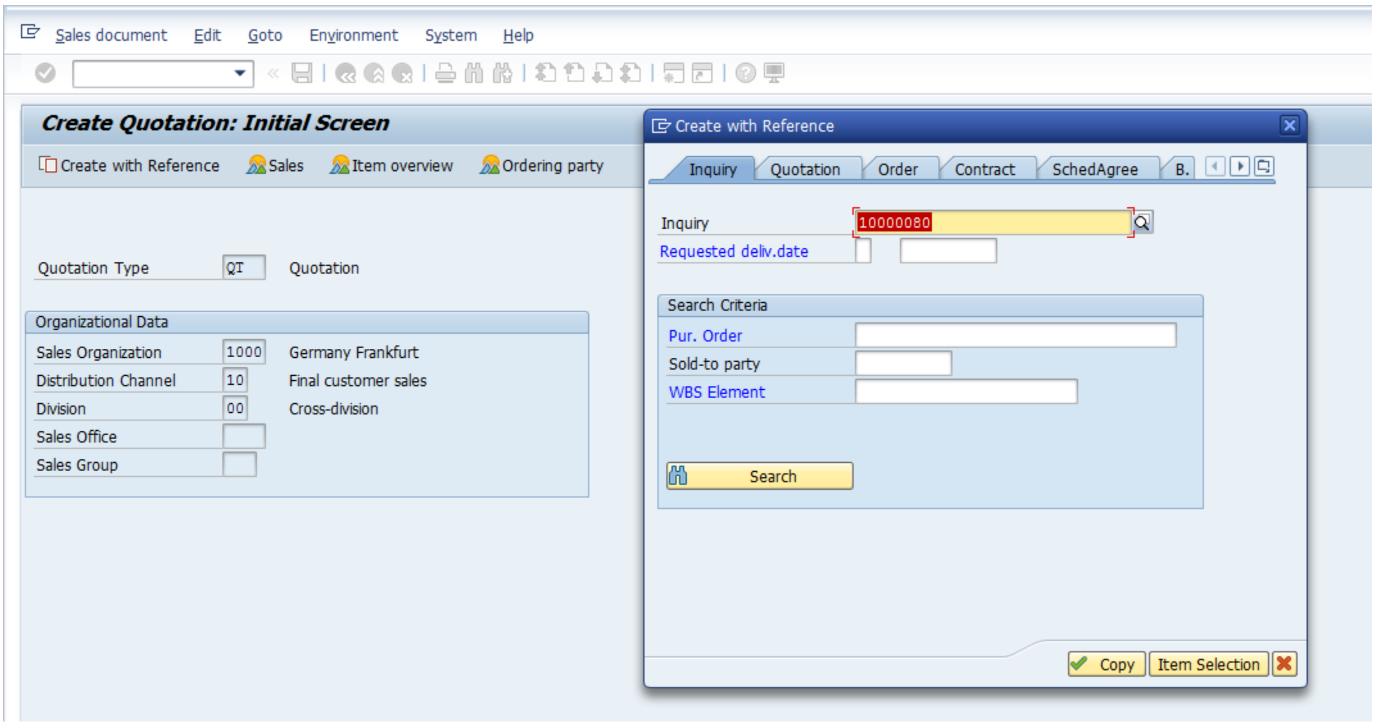
Search Inquiry by Customer

List of ALL Inquiries which are created against the selected customer will be displayed by the SAP system.

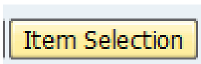


List of Inquiries for a Particular Customer

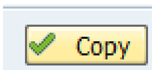
Select the desired inquiry.



Select Inquiry

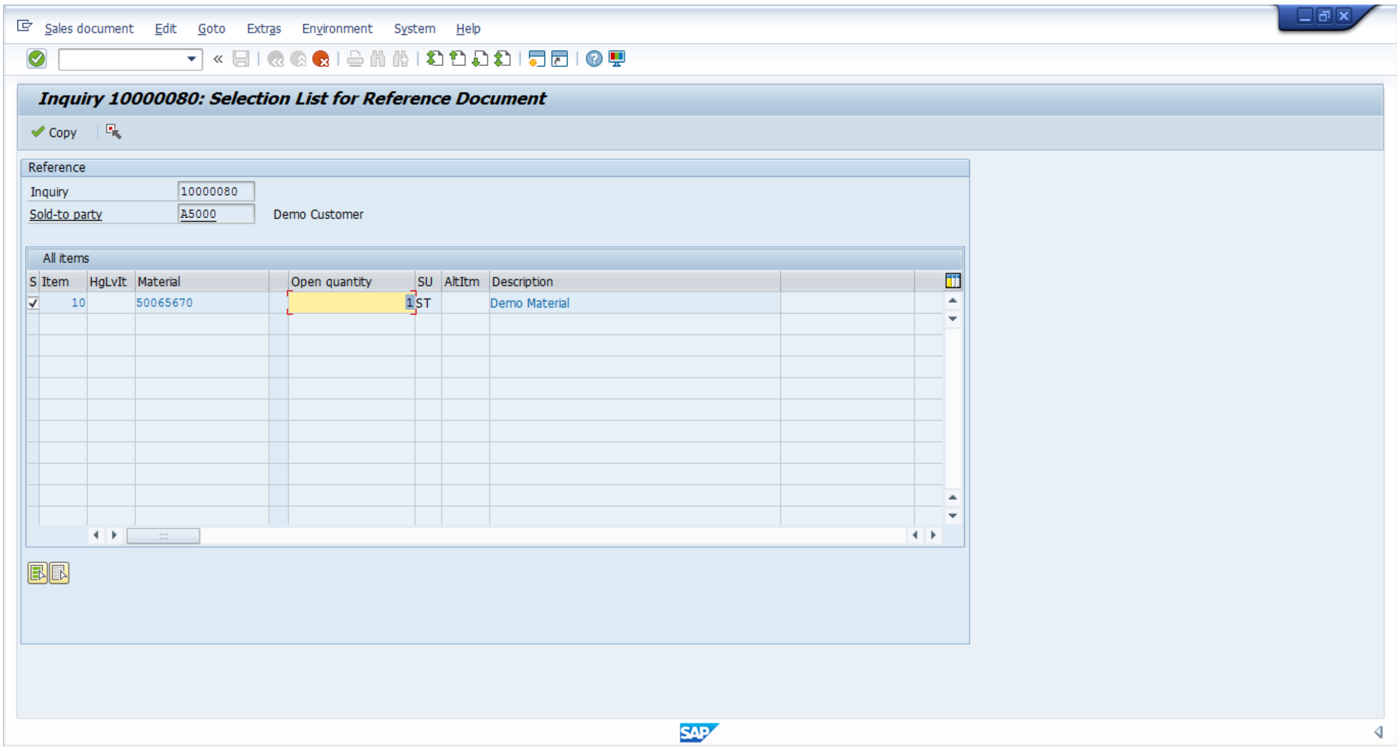
Press either 

, if there are multiple items in the referencing document and you want to select a particular item which should be copied in the quotation or

select 

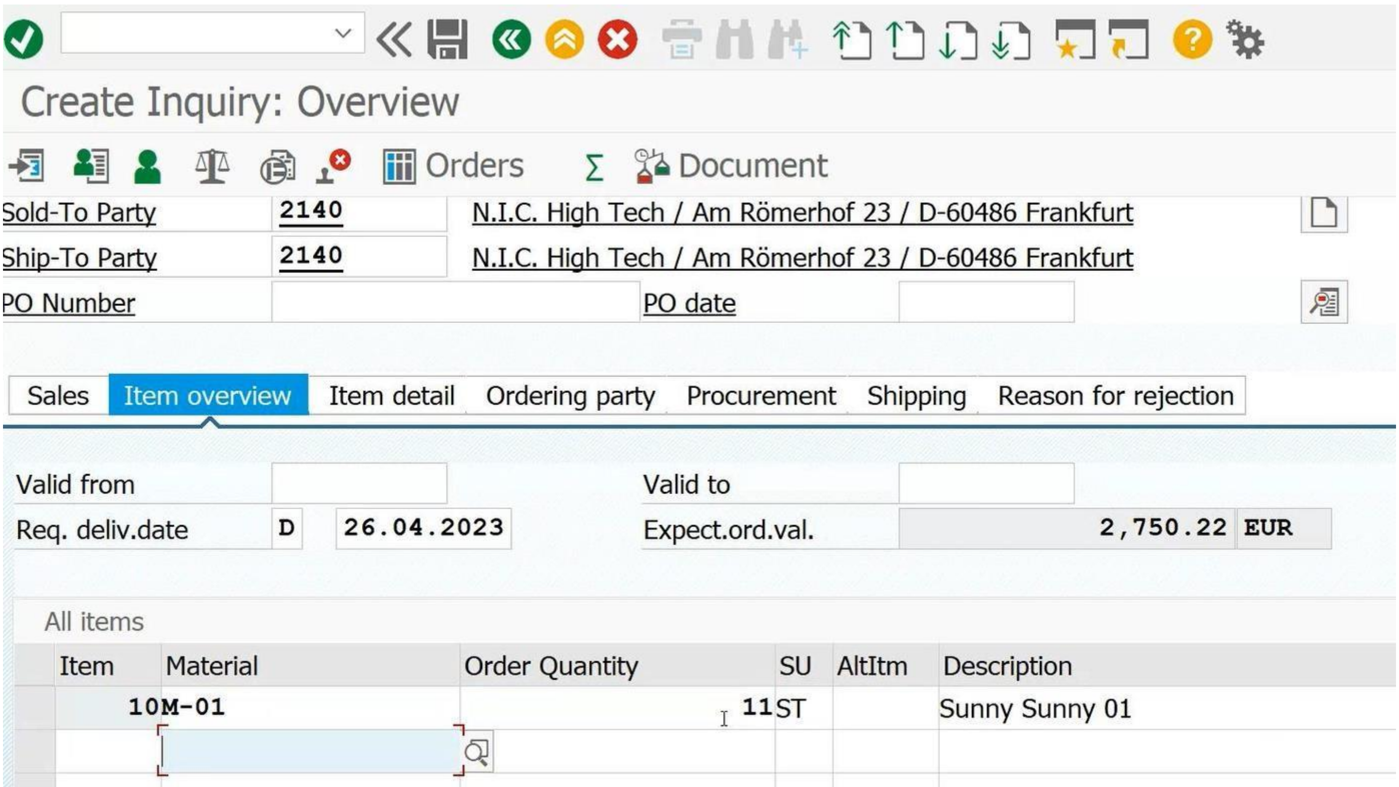
to copy everything directly.

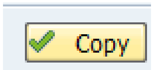
Here we shall press Item Selection button.



Select Items to Copy from Inquiry to Quotation

If there are multiple line items, check box S ☑ can be used to select or deselect an item which is not required to be copied into quotation from the inquiry. Furthermore, you can also edit the quantity if you want to change it from the referencing document.



you Press 

or *F5*, you will enter into the quotation screen.

SAP SD Quotation

You can see that all the information from the Inquiry was copied into the quotation.

8.2 CODDING:

```
REPORT ZPRG_ALV2.
** TO CREATE IMPLICIT WA
TABLES:VBAK,VBAP,SSCRFIELDS.
** TO CREATE VBAK_TY WITH SOME FIELDS OF VBAK
TYPES:BEGIN OF VBAK_TYPE,
    VBELN TYPE VBAK-VBELN,
    TRVOG TYPE VBAK-TRVOG,
    KTEXT TYPE VBAK-KTEXT,
    AUART TYPE VBAK-AUART,
    KUNNR TYPE VBAK-KUNNR,
    ERDAT TYPE VBAK-ERDAT,
    AUDAT TYPE VBAK-AUDAT,
    BSTNK TYPE VBAK-BSTNK,
    ERNAM TYPE VBAK-ERNAM,
    VKORG TYPE VBAK-VKORG,
    VTWEG TYPE VBAK-VTWEG,
    SPART TYPE VBAK-SPART,
    VKGRP TYPE VBAK-VKGRP,
    VKBUR TYPE VBAK-VKBUR,
    WAERK TYPE VBAK-WAERK,
    NETWR TYPE VBAK-NETWR,
    AWAHR TYPE VBAK-AWAHR,
    GUEBG TYPE VBAK-GUEBG,
    GUEEN TYPE VBAK-GUEEN,
    END OF VBAK_TYPE.
** TO CREATE VBAP_TY WITH SOME FIELDS OF VBAP
TYPES:BEGIN OF VBAP_TYPE,
     VBELN  TYPE VBAP-VBELN,
     POSNR  TYPE VBAP-POSNR,
     MATNR  TYPE VBAP-MATNR,
     NETWR TYPE VBAP-NETWR,
     ARKTX  TYPE VBAP-ARKTX,
     KWMENG TYPE VBAP-KWMENG,
```

```
      NETPR  TYPE VBAP-NETPR,
     END OF VBAP_TYPE.
TYPES:BEGIN OF FINAL_TY,
      VBELN TYPE VBAK-VBELN,
      TRVOG TYPE VBAK-TRVOG,
      KTEXT TYPE VBAK-KTEXT,
      AUART TYPE VBAK-AUART,
      KUNNR TYPE VBAK-KUNNR,
      ERDAT TYPE VBAK-ERDAT,
      AUDAT TYPE VBAK-AUDAT,
      BSTNK TYPE VBAK-BSTNK,
      ERNAM TYPE VBAK-ERNAM,
      VKORG TYPE VBAK-VKORG,
      VTWEG TYPE VBAK-VTWEG,
      SPART TYPE VBAK-SPART,
      VKGRP TYPE VBAK-VKGRP,
      VKBUR TYPE VBAK-VKBUR,
      WAERK TYPE VBAK-WAERK,
      NETWR TYPE VBAK-NETWR,
      AWAHR TYPE VBAK-AWAHR,
      GUEBG TYPE VBAK-GUEBG,
      GUEEN TYPE VBAK-GUEEN,
      POSNR  TYPE VBAP-POSNR,
      MATNR  TYPE VBAP-MATNR,
      ARKTX  TYPE VBAP-ARKTX,
      KWMENG TYPE VBAP-KWMENG,
      NETPR  TYPE VBAP-NETPR,
     END OF FINAL_TY.
** to create internal table and wa for both vbak_ty and vbaP_ty
DATA:IT_VBAK TYPE TABLE OF VBAK_TYPE,
    WA_VBAK TYPE VBAK_TYPE,
    IT_VBAP TYPE TABLE OF VBAP_TYPE,
    WA_VBAP TYPE VBAP_TYPE,
    IT_FINAL TYPE TABLE OF FINAL_TY,
    WA_FINAL TYPE FINAL_TY.
** TO CREATE INTERNAL TABLE AND WA FOR FIELDCAT
DATA:IT_FCAT TYPE TABLE OF SLIS_FIELDCAT_ALV,
```

```
    WA_FCAT TYPE SLIS_FIELDCAT_ALV.
DATA:IT_SORT TYPE SLIS_T_SORTINFO_ALV,
    WA_SORT TYPE SLIS_SORTINFO_ALV.
SELECTION-SCREEN BEGIN OF BLOCK BLK1 WITH FRAME TITLE TIT1.
** TO PROVIDE RANGE OF INPUT FIELDS ONTO SELECTION-SCREEN
SELECT-OPTIONS: S_VBELN FOR VBAK-VBELN.
SELECTION-SCREEN SKIP 3.
SELECTION-SCREEN PUSHBUTTON:10(10) PUSH1 USER-COMMAND FCT1,
                    40(10) PUSH2 USER-COMMAND FCT2.

SELECTION-SCREEN END OF BLOCK BLK1.
SELECTION-SCREEN FUNCTION KEY:1,2,3.
INITIALIZATION.
S_VBELN-LOW = '20000172'.
S_VBELN-HIGH = '20000173'.
APPEND S_VBELN.
TIT1 = 'SALES INFO'.
PUSH1 = 'DISPLAY'.
PUSH2 = 'EXIT'.
SSCRFIELDS-FUNCTXT_01 = 'PRINT'.
SSCRFIELDS-FUNCTXT_02 = 'INITIALIZE'.
SSCRFIELDS-FUNCTXT_03 = 'LEAVE'.

AT SELECTION-SCREEN.
IF SSCRFIELDS-UCOMM = 'FCT1'.
SSCRFIELDS-UCOMM = 'ONLI'.
ELSEIF SSCRFIELDS-UCOMM = 'FCT2'.
LEAVE PROGRAM.
ENDIF.

START-OF-SELECTION.
PERFORM VBAK_DATA.
PERFORM VBAP_DATA.
PERFORM BUILD_FCAT.
PERFORM SUB_TOTAL_CALC.
PERFORM DISPLAY_DATA.
```

```
*&---------------------------------------------------------------------*
*&      Form  VBAK_DATA
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  --> p1        text
*  <-- p2        text
*----------------------------------------------------------------------*
FORM VBAK_DATA .
SELECT VBELN TRVOG KTEXT AUART KUNNR ERDAT AUDAT BSTNK ERNAM VKORG VTW
EG SPART VKGRP VKBUR WAERK NETWR AWAHR GUEBG GUEEN FROM VBAK INTO TABLE
IT_VBAK WHERE
VBELN IN S_VBELN.
ENDFORM.
*&---------------------------------------------------------------------*
*&      Form  VBAP_DATA
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  --> p1        text
*  <-- p2        text
*----------------------------------------------------------------------*
FORM VBAP_DATA .
IF IT_VBAK IS NOT INITIAL.
SELECT VBELN POSNR MATNR NETWR ARKTX KWMENG NETPR FROM VBAP INTO TABLE I
T_VBAP FOR ALL ENTRIES IN IT_VBAK WHERE VBELN = IT_VBAK-
VBELN AND NETWR = IT_VBAK-NETWR.
ENDIF.
LOOP AT IT_VBAP INTO WA_VBAP.
WA_FINAL-POSNR = WA_VBAP-POSNR.
WA_FINAL-MATNR = WA_VBAP-MATNR.
WA_FINAL-ARKTX = WA_VBAP-ARKTX.
WA_FINAL-KWMENG = WA_VBAP-KWMENG.
WA_FINAL-NETPR = WA_VBAP-NETPR.
READ TABLE IT_VBAK INTO WA_VBAK WITH KEY VBELN = WA_VBAP-VBELN.
IF SY-SUBRC = 0.
WA_FINAL-VBELN = WA_VBAK-VBELN.
```

```
WA_FINAL-TRVOG = WA_VBAK-TRVOG.
WA_FINAL-KTEXT = WA_VBAK-KTEXT.
WA_FINAL-AUART = WA_VBAK-AUART.
WA_FINAL-KUNNR = WA_VBAK-KUNNR.
WA_FINAL-ERDAT = WA_VBAK-ERDAT.
WA_FINAL-AUDAT = WA_VBAK-AUDAT.
WA_FINAL-BSTNK = WA_VBAK-BSTNK.
WA_FINAL-ERNAM = WA_VBAK-ERNAM.
WA_FINAL-VKORG = WA_VBAK-VKORG.
WA_FINAL-VTWEG = WA_VBAK-VTWEG.
WA_FINAL-SPART = WA_VBAK-SPART.
WA_FINAL-VKGRP = WA_VBAK-VKGRP.
WA_FINAL-VKBUR = WA_VBAK-VKBUR.
WA_FINAL-WAERK = WA_VBAK-WAERK.
WA_FINAL-NETWR = WA_VBAK-NETWR.
WA_FINAL-AWAHR = WA_VBAK-AWAHR.
WA_FINAL-GUEBG = WA_VBAK-GUEBG.
WA_FINAL-GUEEN = WA_FINAL-GUEEN.
ENDIF.
APPEND WA_FINAL TO IT_FINAL.
CLEAR:WA_FINAL,WA_VBAK,WA_VBAP.
ENDLOOP.
ENDFORM.
*&---------------------------------------------------------------------*
*&      Form  BUILD_FCAT
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  --> p1        text
*  <-- p2        text
*----------------------------------------------------------------------*
FORM BUILD_FCAT .
REFRESH IT_FCAT.
WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 1.
WA_FCAT-FIELDNAME = 'VBELN'.
WA_FCAT-TABNAME = 'IT_VBAK'.
```

```
WA_FCAT-JUST = 'L'.
WA_FCAT-OUTPUTLEN = 20.
WA_FCAT-SELTEXT_S = 'VBELN'.
WA_FCAT-SELTEXT_M = 'SALES DOC'.
WA_FCAT-SELTEXT_L = ' SALES DOCUMENT'.
WA_FCAT-NO_ZERO = 'X'.
WA_FCAT-NO_CONVEXT = 'X'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 2.
WA_FCAT-FIELDNAME = 'TRVOG'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'TRVOG'.
WA_FCAT-SELTEXT_M = 'TRANSACTION GRP'.
WA_FCAT-SELTEXT_L = ' TRANSACTION GROUP'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 3.
WA_FCAT-FIELDNAME = 'KTEXT'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'KTEXT'.
WA_FCAT-SELTEXT_M = 'DESCRIPTION'.
WA_FCAT-SELTEXT_L = ' DESCRIPTION'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 4.
WA_FCAT-FIELDNAME = 'AUART'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'R'.
```

```
WA_FCAT-SELTEXT_S = 'AUART'.
WA_FCAT-SELTEXT_M = 'SALES DOC TYP'.
WA_FCAT-SELTEXT_L = ' SALES DOCUMENT TYPE'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 5.
WA_FCAT-FIELDNAME = 'KUNNR'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'KUNNR'.
WA_FCAT-SELTEXT_M = 'SOLD PARTY'.
WA_FCAT-SELTEXT_L = ' SOLD TO PARTY'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-EDIT_MASK   =   '__/__/____'.
WA_FCAT-COL_POS = 6.
WA_FCAT-FIELDNAME = 'ERDAT'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'ERDAT'.
WA_FCAT-SELTEXT_M = 'CREATED ON'.
WA_FCAT-SELTEXT_L = ' CREATED ON '.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-EDIT_MASK   =   '__/__/____'.
WA_FCAT-COL_POS = 7.
WA_FCAT-FIELDNAME = 'AUDAT'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'AUDAT'.
WA_FCAT-SELTEXT_M = 'DOC DATE'.
```

```
WA_FCAT-SELTEXT_L = ' DOCUMENT DATE'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 8.
WA_FCAT-FIELDNAME = 'BSTNK'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'PO No'.
WA_FCAT-SELTEXT_M = 'PURCHESE ORDER NUM'.
WA_FCAT-SELTEXT_L = 'PURCHESE ORDER NUMBER'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 9.
WA_FCAT-FIELDNAME = 'ERNAM'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'ERNAM'.
WA_FCAT-SELTEXT_M = 'CREATED BY'.
WA_FCAT-SELTEXT_L = ' CREATED BY'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 10.
WA_FCAT-FIELDNAME = 'VKORG'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'VKORG'.
WA_FCAT-SELTEXT_M = 'SALES ORG'.
WA_FCAT-SELTEXT_L = ' SALES ORGANIZATION'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.
```

```
WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 11.
WA_FCAT-FIELDNAME = 'VTWEG'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'VTWEG'.
WA_FCAT-SELTEXT_M = 'DISTRIBUTION'.
WA_FCAT-SELTEXT_L = ' DISTRIBUTION CHANNEL'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 12.
WA_FCAT-FIELDNAME = 'SPART'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'SPART'.
WA_FCAT-SELTEXT_M = 'DIVISION'.
WA_FCAT-SELTEXT_L = ' DIVISION'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 13.
WA_FCAT-FIELDNAME = 'VKGRP'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'VKGRP'.
WA_FCAT-SELTEXT_M = 'SALES OFC'.
WA_FCAT-SELTEXT_L = 'SALES OFFICE'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 14.
WA_FCAT-FIELDNAME = 'VKBUR'.
WA_FCAT-TABNAME = 'IT_VBAK'.
```

```
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'VKBUR'.
WA_FCAT-SELTEXT_M = 'SALES GRP'.
WA_FCAT-SELTEXT_L = ' SALES GROUP'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 15.
WA_FCAT-FIELDNAME = 'WAERK'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'WAERK'.
WA_FCAT-SELTEXT_M = 'DOCUMENT CNY'.
WA_FCAT-SELTEXT_L = ' DOCUMENT CURRENCY'.
WA_FCAT-CURRENCY = 'X'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 16.
WA_FCAT-FIELDNAME = 'NETWR'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'NETWR'.
WA_FCAT-SELTEXT_M = 'NET VALUE'.
WA_FCAT-SELTEXT_L = 'NET VALUE'.
WA_FCAT-DO_SUM = 'X'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 17.
WA_FCAT-FIELDNAME = 'AWAHR'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'AWAHR'.
```

```
WA_FCAT-SELTEXT_M = 'PROBABILITY'.
WA_FCAT-SELTEXT_L = ' PROBABILITY'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-EDIT_MASK   =   '__/__/____'.
WA_FCAT-COL_POS = 18.
WA_FCAT-FIELDNAME = 'GUEBG'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'GUEBG'.
WA_FCAT-SELTEXT_M = 'VALID FROM'.
WA_FCAT-SELTEXT_L = ' VALID FROM'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-EDIT_MASK   =   '__/__/____'.
WA_FCAT-COL_POS = 19.
WA_FCAT-FIELDNAME = 'GUEEN'.
WA_FCAT-TABNAME = 'IT_VBAK'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'GUEEN'.
WA_FCAT-SELTEXT_M = 'VALID UPTO'.
WA_FCAT-SELTEXT_L = 'VALID UPTO DATE'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 20.
WA_FCAT-FIELDNAME = 'POSNR'.
WA_FCAT-TABNAME = 'IT_VBAP'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'POSNR'.
WA_FCAT-SELTEXT_M = 'SALES DOC ITEM'.
WA_FCAT-SELTEXT_L = 'SALES DOCUMENT ITEM'.
```

```
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.

WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 21.
WA_FCAT-FIELDNAME = 'MATNR'.
WA_FCAT-TABNAME = 'IT_VBAP'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'MATNR'.
WA_FCAT-SELTEXT_M = 'MATERIAL NO'.
WA_FCAT-SELTEXT_L = 'MATERIAL NUMBER'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.


WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 22.
WA_FCAT-FIELDNAME = 'ARKTX'.
WA_FCAT-TABNAME = 'IT_VBAP'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'ARKTX'.
WA_FCAT-SELTEXT_M = 'SALES ORDER'.
WA_FCAT-SELTEXT_L = 'SALES ORDER ITAM'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.


WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 23.
WA_FCAT-FIELDNAME = 'KWMENG'.
WA_FCAT-TABNAME = 'IT_VBAP'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'QTY'.
WA_FCAT-SELTEXT_M = 'ORDERED QTY'.
WA_FCAT-SELTEXT_L = 'ORDERED QUANTITY'.
*WA_FCAT-DO_SUM = 'X'.
APPEND WA_FCAT TO IT_FCAT.
```

```
CLEAR WA_FCAT.



WA_FCAT-ROW_POS  = 1.
WA_FCAT-COL_POS = 24.
WA_FCAT-FIELDNAME = 'NETPR'.
WA_FCAT-TABNAME = 'IT_VBAP'.
WA_FCAT-JUST = 'L'.
WA_FCAT-SELTEXT_S = 'NETPR'.
WA_FCAT-SELTEXT_M = 'NET PRICE'.
WA_FCAT-SELTEXT_L = 'NET PRICE'.
WA_FCAT-DO_SUM = 'X'.
APPEND WA_FCAT TO IT_FCAT.
CLEAR WA_FCAT.
ENDFORM.

*&---------------------------------------------------------------------*
*&      Form  SUB_TOTAL_CALC
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  --> p1        text
*  <-- p2        text
*----------------------------------------------------------------------*
FORM SUB_TOTAL_CALC .
WA_SORT-FIELDNAME = 'VBELN'.
WA_SORT-UP = 'X'.
WA_SORT-SUBTOT = 'X'.
APPEND WA_SORT TO IT_SORT.
CLEAR WA_SORT.
ENDFORM.
*&---------------------------------------------------------------------*
*&      Form  DISPLAY_DATA
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  --> p1        text
```

```
*  <-- p2       text
*----------------------------------------------------------------------*
FORM DISPLAY_DATA .
CALL FUNCTION 'REUSE_ALV_GRID_DISPLAY'
 EXPORTING
   I_CALLBACK_PROGRAM              = SY-REPID
   I_CALLBACK_PF_STATUS_SET        = 'HANDLE_GUI'
*  I_CALLBACK_USER_COMMAND         = ' '
*  I_CALLBACK_TOP_OF_PAGE          = ' '
*  I_CALLBACK_HTML_TOP_OF_PAGE     = ' '
*  I_CALLBACK_HTML_END_OF_LIST     = ' '
*  I_STRUCTURE_NAME                =
*  I_BACKGROUND_ID                 = ' '
   I_GRID_TITLE                    = 'QUOTATION FOR MATERIAL REQUISITION'
*  I_GRID_SETTINGS                 =
*  IS_LAYOUT                       =
   IT_FIELDCAT                     = IT_FCAT
*  IT_EXCLUDING                    =
*  IT_SPECIAL_GROUPS               =
   IT_SORT                         = IT_SORT
*  IT_FILTER                       =
*  IS_SEL_HIDE                     =
*  I_DEFAULT                       = 'X'
*  I_SAVE                          = ' '
*  IS_VARIANT                      =
*  IT_EVENTS                       =
*  IT_EVENT_EXIT                   =
*  IS_PRINT                        =
*  IS_REPREP_ID                    =
*  I_SCREEN_START_COLUMN           = 0
*  I_SCREEN_START_LINE             = 0
*  I_SCREEN_END_COLUMN             = 0
*  I_SCREEN_END_LINE               = 0
*  I_HTML_HEIGHT_TOP               = 0
*  I_HTML_HEIGHT_END               = 0
*  IT_ALV_GRAPHICS                 =
*  IT_HYPERLINK                    =
```

```
*   IT_ADD_FIELDCAT              =
*   IT_EXCEPT_QINFO              =
*   IR_SALV_FULLSCREEN_ADAPTER        =
* IMPORTING
*   E_EXIT_CAUSED_BY_CALLER           =
*   ES_EXIT_CAUSED_BY_USER            =
 TABLES
   T_OUTTAB                    = IT_FINAL
 EXCEPTIONS
   PROGRAM_ERROR                 = 1
   OTHERS                   = 2
        .
IF SY-SUBRC <> 0.
MESSAGE 'DOCUMENT NOT FOUND' TYPE 'E'.
ENDIF.

ENDFORM.

 FORM HANDLE_GUI USING IT_FCODES TYPE SLIS_T_EXTAB.
   SET PF-STATUS 'ALV_GUI'.

   ENDFORM.
```

9. TESTING

9. TESTING

Introduction to testing

Testing is the process of evaluating a system or its component(s) with the intent to find whether it satisfies the specified requirements or not. Testing is executing a system in order to identify any gaps, errors, or missing requirements in contrary to the actual requirements. According to ANSI/IEEE 1059 standard, Testing can be defined as - A process of analyzing a software item to detect the differences between existing and required conditions (that is defects/errors/bugs) and to evaluate the features of the software item.

Testing in SAP ABAP is an important process that ensures the quality and reliability of the software. Here are some notes on testing in SAP ABAP.

1. **Types of Testing**: There are several types of testing that can be performed in SAP ABAP, including unit testing, integration testing, system testing, and acceptance testing. Each type of testing has a different purpose and should be performed at different stages of the development process.
2. **Unit Testing**: Unit testing is the process of testing individual units of code to ensure that they work as expected. This is typically performed by the developer and should be done after each unit of code is completed.

3. **Integration Testing**: Integration testing is the process of testing how individual units of code work together as a system. This is typically performed by a team of developers and should be done after all units of code have been completed.

4. **System Testing**: System testing is the process of testing the entire system to ensure that it meets the requirements and works as expected. This is typically performed by a team of testers and should be done before the system is deployed.

5. **Acceptance Testing**: Acceptance testing is the process of testing the system from the user's perspective to ensure that it meets their needs and works as expected. This is typically performed by the end-users and should be done before the system is deployed.

6. **Tools for Testing**: SAP ABAP provides various tools for testing, including the ABAP Unit testing framework, the ABAP Test Cockpit, and the ABAP Debugger. These tools can help developers and testers to identify and fix issues in the code.

7. **Best Practices**: To ensure successful testing in SAP ABAP, it is important to follow best practices, such as defining clear requirements, creating a testing plan, using test data, and documenting test results.

In summary, testing is an essential part of SAP ABAP development and should be performed at various stages of the development process to ensure the quality and reliability of the software.

Functional Testing

This is a type of black-box testing that is based on the specifications of the software that is to be tested. The application is tested by providing input and then the results are examined that need to conform to the functionality it was intended for. Functional testing of a software is conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements.

Software Testing Life Cycle

The process of testing a software in a well-planned and systematic way is known as softwaretesting lifecycle (STLC).

Different organizations have different phases in STLC however generic Software Test LifeCycle (STLC) for waterfall development model consists of the following phases.

1. Requirements Analysis
2. Test Planning
3. Test Analysis
4. Test Design

Requirements Analysis:

In this phase testers analyze the customer requirements and work with developers during the design phase to see which requirements are testable and how they are going to test those requirements. It is very important to start testing activities from the requirements phase itself because the cost of fixingdefect is very less if it is found in requirements phase rather than in future phases.

Test Planning:

In this phase all the planning about testing is done like what needs to be tested, how the testing will be done, test strategy to be followed, what will be the test environment, what test methodologies will be followed, hardware and software availability, resources, risks etc. A high-level test plan document is created which includes all the planning inputs mentioned above and circulated to the stakeholders.

Test Analysis:

After test planning phase is over test analysis phase starts, in this phase we need to dig deeper into project and figure out what testing needs to be carried out in each SDLC phase. Automation activities are also decided in this phase, if automation needs to be done for software product, how will the automation be done, how much time will it taketo automate and which features need to be automated. Nonfunctional testing areas (Stress and performance testing) are also analyzed and defined in this phase.

Test Design:

In this phase various black-box and white-box test design techniques are used to design the test cases for testing, testers start writing test cases by following those design techniques, if automation testing needsto be done then automation scripts also need to written in this phase.

Test Cases

1. The model is tested with test data.
2. Accuracy is calculated for each algorithm.

10. RESULTS

10. RESULTS

11. CONCLUSION

11. CONCLUSION

The quotation is the formal document that outlines the proposed solution and pricing to a potential customer, based on their inquiry or request.

This process requires clear communication between the sales team and the customer to fully understand their requirements and expectations. The sales team must provide accurate information about the products or services offered and create a tailored solution that meets the customer's needs.

Creating a professional and detailed quotation can help to build trust and credibility with the customer, and increase the chances of closing the sale. Timely and accurate follow-up is also important to ensure that the customer receives the information they need to make an informed decision.

Overall, a well-executed sales and inquiry quotation process can help businesses to build strong customer relationships, increase sales, and ultimately, achieve their business objectives.

12. FUTURE SCOPE

12. FUTURE SCOPE

The sales and inquiry quotation process are an essential part of any sales cycle, and its future scope is closely tied to the evolution of technology and customer needs. Here are some potential future developments and trends that could impact the sales and inquiry quotation process:

1. Automation: As technology advances, the sales and inquiry quotation process may become more automated, with the use of chatbots and AI to provide quick and accurate responses to customer inquiries.

2. Personalization: With increased access to customer data, businesses will be able to provide more personalized and tailored quotations that take into account the customer's specific needs and preferences.

3. Collaboration: In the future, the sales and inquiry quotation process may involve more collaboration between different departments within an organization, such as marketing, sales, and customer service, to provide a seamless and cohesive customer experience.

4. Mobile: With the growing use of mobile devices, sales and inquiry quotations may become more mobile-friendly, with the ability to view and accept quotations on-the-go.

13. REFERENCES

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We writing to recommend **MOHAMMAD IRFAN SIR** for their excellent support in creating a project for us. We had the pleasure of working with Irfan sir on a project, and we were impressed by his professionalism, expertise, and dedication to delivering a high-quality project.

Throughout the project, Irfan sir demonstrated exceptional technical skills and knowledge, and he was always willing to go the extra mile to ensure that the project was completed on time and to the highest standard. His attention to detail and ability to anticipate and address potential issues ensured that the project was completed smoothly and efficiently.

Irfan sir was also an excellent communicator, keeping us informed at every stage of the project and responding promptly to any questions or concerns we had. His friendly and approachable demeanor made him easy to work with us, and he was always willing to listen and adapt to our needs and preferences.

Overall, I am extremely satisfied with the project that Irfan sir and his support to create for us, and we would highly recommend him to anyone seeking a skilled and reliable professional for their project needs.

Sincerely,
MOHAMMAD BIN SALAM BAJKHAIF (19626T0904),
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